

Turacoz IT Service Desk

Complete Help & Reference Guide

Application **turacoz_service_desk** · Module **Turacoz IT Service Desk**

Version 1.0.1 · ERPNext 15 / Frappe 15

Enterprise ITSM and ticketing, aligned to ITIL practice

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1. How to use this guide

This document is the single reference for the Turacoz IT Service Desk. It is written so that somebody who has never seen the system can read it front to back and then operate, configure, support or extend the application.

The guide is generated from the application source itself — the DocType JSON definitions, the controllers, the engines, the reports and the installation scripts — so every field name, status value, permission, endpoint and schedule below is what the running system actually does, not what it was once intended to do.

If you are...	Read
An employee who just wants to raise a request	Sections 2 and 4.1, plus section 12 (the portal)
A service desk engineer	Sections 2, 4.2, 5 and 9
A team lead or IT manager	Sections 3, 4.3–4.5, 7, 8 and 16
An approver (manager, department head, CISO)	Sections 4.4, 5.5 and 7.3
An auditor	Sections 3, 5, 7.5 and 14
Configuring the system for the first time	Sections 15 and 16, then 7.1–7.3
Building an integration	Sections 6, 11 and 14
Extending or maintaining the code	Sections 6, 7, 14 and 19

Conventions. `Fixed width` is a field name, file path, endpoint or shell command. **Bold** inside a sentence is usually a status value, a role or a button. Where a rule is enforced in code (rather than merely recommended), this guide says so explicitly — those are the rules you cannot work around from the user interface.

2. The system at a glance

2.1 What it does

The Turacoz IT Service Desk is a complete IT service management (ITSM) application for ERPNext 15. It takes every request that reaches the IT department, records it, routes it to the right team and person, holds it to a service level agreement measured in real business hours, escalates it when it is running late, walks it through approval where approval is needed, and keeps an immutable record of everything that happened for audit.

Capability	What you get
Ticketing	Service Ticket with a 12-state flow, work logs, public and internal comments, watchers, satisfaction feedback, attachments and related records.
ITIL practices	Incident, Problem, Change Request (with CAB, risk rating and rollback plan) and Privileged Access Request with automatic expiry.
SLA	Per-priority response and resolution targets computed in working time, holiday-aware, with pause/resume, breach detection and a multi-level escalation ladder.
Assignment	Round Robin, Load Balancing, Skill Based or Manual, mirrored into Frappe's own ToDo so it shows in the assignment sidebar.
Approvals	Approval Matrix with role, specific user, department head, team lead, IT manager, CISO or requester's manager levels; sequential or parallel; optional python conditions.
Knowledge base	Versioned articles with approval, keyword auto-suggestion against ticket text, and one-click article creation from a resolved ticket.
CMDB	Configuration Items with typed relationships, environment, criticality and an optional link to an ERPNext Asset.
Audit	Append-only Service Desk Audit Log capturing user, IP, user agent and a JSON field-level diff, with a 7-year default retention.
Analytics	12 script reports, 8 number cards, 5 dashboard charts and 4 role-specific dashboard APIs.
Self service	An employee portal at <code>/support</code> .
Integration	Token-authenticated REST endpoints returning a consistent JSON envelope.

2.2 Where things live

Place	URL / path	Who uses it
Workspace	<code>/app/it-service-desk</code>	Everyone with desk access
Ticket list	<code>/app/service-ticket</code>	Engineers, leads, executives
Employee portal	<code>/support</code>	All employees
Settings	<code>/app/service-desk-settings</code>	System Manager, IT Manager
Audit log	<code>/app/service-desk-audit-log</code>	System Manager, IT Manager, CISO, Auditor

Place	URL / path	Who uses it
Reports	/app/query-report/<report name>	By role, see section 8
REST API	/api/method/ turacoz_service_desk.turacoz_it_service_desk.api.<module>.<method>	Integrations
Error log	/app/error-log	Support — entries are titled "Service desk..."

2.3 Vocabulary

Six words carry most of the meaning in this system. Getting them right prevents almost every misunderstanding.

Term	Means
Ticket	Anything a user asks IT for. The container everything else hangs off.
Incident	A service is broken or degraded right now. Focus is restoration.
Problem	The cause behind repeated incidents. Focus is permanent elimination.
Change	A deliberate modification to a live service. Focus is controlled risk.
Impact × Urgency = Priority	Impact is how many people or how much business is affected; urgency is how fast it must move. The system multiplies them into a priority.
SLA	Two promises: how fast somebody will respond, and how fast it will be resolved. Both are measured in working hours.

The impact × urgency matrix is fixed in code and applied automatically unless Derive Priority From Impact & Urgency is switched off in Service Desk Settings, or unless a user has explicitly changed the priority on this save.

Impact \ Urgency	High	Medium	Low
High	Critical	High	Medium
Medium	High	Medium	Low
Low	Medium	Low	Low

3. Roles and who can do what

3.1 The nine roles

Seven roles are created by the installer; two more (**Employee** and **System Manager**) already exist in every ERPNext site and are reused. Existing roles are never overwritten.

Role	Purpose	Typically held by
Employee	Raise a request, comment on it, verify the fix, give feedback. Sees only their own tickets.	Every member of staff
Service Desk Executive	Triage, assign and monitor tickets. Full read across the desk, can create and edit tickets, comments, work logs and knowledge articles.	Front-line service desk staff
Service Desk Engineer	Works assigned tickets, logs effort, resolves. Can create incidents, problems, changes and knowledge articles.	Support engineers
Service Desk Team Lead	Owens a team, reassigns work, receives the first escalation rung, reviews SLA. Can maintain teams, skills, categories and the catalog.	Team leaders
Department Head	Approves requests raised by their department. Read-only on service desk records.	Line-of-business heads
IT Manager	Full control over the service desk: settings, SLA policies, approval matrices, all reports, all approvals.	IT management
CISO	Security incidents, privileged access approval and audit oversight. Writes on Incident and Privileged Access Request; reads everything else.	Security officer
Auditor	Read-only everywhere, including the audit log, with report and export rights. Cannot create, edit or delete anything.	Internal and external audit
System Manager	Full administrative control, including delete.	ERPNext administrators

3.2 The ticket visibility rule

Service Ticket carries a custom permission rule — registered in `hooks.py` as both a `permission_query_conditions` (which filters lists and reports) and a `has_permission` check (which guards individual documents).

The rule. If the user holds any of **System Manager**, **Administrator**, **IT Manager**, **CISO**, **Auditor**, **Service Desk Executive**, **Service Desk Engineer** or **Service Desk Team Lead**, they see **every** ticket.

Otherwise they see only tickets where they are the `requester`, the `requested_for`, the document `owner`, or the `assigned_engineer`.

This is why an employee who complains that they can see colleagues' tickets is almost always holding a service desk role they should not have. Visibility is granted by role first, then by relationship — there is no third mechanism.

3.3 Full permission matrix

Rights are abbreviated: **R** read · **W** write · **C** create · **D** delete · **Y** report · **E** export · **H** share · **P** print · **M** e-mail. So **RWCDYEHMP** is full control and **RYEPM** is read-only with reporting. The same abbreviations are used in the per-DocType permission tables in section 6. Child tables are omitted — they inherit the permissions of their parent document.

Column keys: **SM** System Manager · **ITM** IT Manager · **CISO** CISO · **TL** Service Desk Team Lead · **EXE** Service Desk Executive · **ENG** Service Desk Engineer · **DH** Department Head · **EMP** Employee · **AUD** Auditor.

DocType	SM	ITM	CISO	TL	EXE	ENG	DH	EMP	AUD
Approval Matrix	RWCDYEHMP	RWCDYEHMP	RYPM	RYPM	–	–	RYPM	–	RYEPM
Change Request	RWCDYEHMP	RWCDYEHMP	RYPM	RWCYHPM	RYPM	RWCYHPM	RYPM	–	RYEPM
Configuration Item	RWCDYEHMP	RWCDYEHMP	RYPM	RWCYHPM	RYPM	RWCYHPM	–	–	RYEPM
Engineer Skill	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RYPM	RYPM	–	–	RYEPM
IT Service Category	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RYPM	RYPM	–	RYPM	RYEPM
Incident	RWCDYEHMP	RWCDYEHMP	RWCYHPM	RWCYHPM	RYPM	RWCYHPM	–	–	RYEPM
Knowledge Article	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RWCYHPM	RWCYHPM	–	RYPM	RYEPM
Privileged Access Request	RWCDYEHMP	RWCDYEHMP	RWCDYEHMP	RWCYHPM	–	RYPM	RYPM	RWCYPM	RYEPM
Problem	RWCDYEHMP	RWCDYEHMP	RYPM	RWCYHPM	RYPM	RWCYHPM	–	–	RYEPM
SLA Policy	RWCDYEHMP	RWCDYEHMP	–	RYPM	RYPM	RYPM	–	–	RYEPM
Service Catalog	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RYPM	RYPM	–	RYPM	RYEPM
Service Desk Audit Log	RYEPM	RYEPM	RYEPM	–	–	–	–	–	RYEPM
Service Desk Settings	RWCDHMP	RWCDHMP	–	–	–	–	–	–	RPM
Service Desk Team	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RYPM	RYPM	–	–	RYEPM
Service Ticket	RWCDYEHMP	RWCDYEHMP	RYPM	RWCYHPM	RWCYHPM	RWCYHPM	RYPM	RWCYHPM	RYEPM
Ticket Comment	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RWCYHPM	RWCYHPM	–	RWCYPM	RYEPM
Ticket Feedback	RWCDYEHMP	RWCDYEHMP	–	RYPM	RYPM	RYPM	–	RWCYPM	RYEPM
Work Log	RWCDYEHMP	RWCDYEHMP	–	RWCYHPM	RWCYHPM	RWCYHPM	–	–	RYEPM

4. Day-to-day working

This section is the "what do I actually do" chapter. Each sub-section follows one role through a normal working day.

4.1 The employee — raising and tracking a request

1. Go to `/support` (or use the **IT Service Desk** entry in the portal menu at `/me`). You must be logged in; guests are refused.
2. Choose a **service** from the catalog if one fits (New Laptop, VPN Access, Password Reset, ...), otherwise pick a **category**. Picking a service pre-fills the category, sub-category, priority and department for you.
3. Type a subject and a description. As you type, the system suggests **knowledge articles** that match your words — read them first; many requests are answered without a ticket.
4. Submit. You get a ticket number in the form `TKT-2026-000123` and, if the service does not need approval, an engineer is assigned immediately.
5. Track it under **My Tickets**. Two counters matter: **open**, and **awaiting me** — the latter means the ticket is in **Pending User** or **User Verification** and is waiting on you, not on IT. While a ticket sits in Pending User the SLA clock is paused.
6. When the engineer marks it **Resolved**, you are e-mailed and asked to verify. Confirm and it moves to **Closed**; if it is not actually fixed, reopen it — that restarts the resolution clock and is counted against the reopen rate.
7. If you do nothing, the daily job auto-closes the ticket after the configured number of days (default 3).
8. After closure you are sent a short satisfaction survey. One rating per ticket.

4.2 The engineer — working the queue

1. Open the workspace or your engineer dashboard. Your queue is sorted **Critical → High → Medium → Low**, then by the nearest resolution deadline.
2. The list view colours each row by status, and shows a red **SLA Breached** pill for any open ticket whose SLA status is Response Overdue, Resolution Overdue or Failed. Deal with those first.
3. Open a ticket. The form shows live SLA countdowns for both the response and resolution clocks, the percentage of each target consumed, total hold time and the escalation level.
4. **Respond**. Your first comment, first work log or first move into a work status stamps the first response and settles the response half of the SLA. Do this early — it is measured.
5. Use **Add Work Log** to record effort. Start and end time produce the hours; the total rolls up onto the ticket and into your performance report.
6. Use comments deliberately. A **Public Note** is e-mailed to the requester and is visible on their portal. An **Internal Note** is visible only to service desk roles.
7. If you are waiting on the user or a vendor, move the ticket to **Pending User** or **Pending Vendor**. That pauses the SLA clock, and the held time is added back to both deadlines when you resume — so use it honestly rather than to hide a slipping ticket.
8. Use **Suggest Articles** before you start digging. Use **Create → Knowledge Article** after you finish so the next person does not repeat your work.
9. If the ticket reveals a broken service, a recurring cause or a needed change, use **Create → Incident / Problem / Change Request**. The new record is linked back to the ticket in both directions.
10. To resolve, you **must** fill in the **Resolution** field — the save is blocked otherwise. Set the resolution category (it defaults to **Fixed**).

4.3 The team lead

- **Escalations land on you first.** The seeded SLA policy escalates to Service Desk Team Lead at 75 % of both the response and the resolution target.
- **Reassign** from the ticket form. You can move a ticket to another engineer, another team, or both; a reason is recorded in the assignment history and the timeline.
- **Maintain the team.** Add and remove members, set each member's Max Open Tickets, mark people unavailable when they are on leave (this removes them from automatic assignment entirely), and choose the assignment method.
- **Maintain skills** if you use Skill Based assignment — without Engineer Skill rows it silently falls back to Load Balancing.
- **Watch the reports:** Ticket Aging for the backlog shape, SLA Violations for what broke, Reopened Tickets for quality, Engineer Performance for the team.

4.4 The approver

- You are told by e-mail and by an in-app alert when something needs you.
- Your inbox of pending items across tickets, changes and access requests comes from `api.itsm.pending_approvals`, and appears on the employee dashboard as `awaiting_my_approval`.
- Open the document. **Approve** and **Reject** buttons appear only if you are genuinely a pending approver at the current level — the form asks the server before showing them.
- **A rejection needs a reason.** The system refuses to reject without comments.
- Approving moves the document to the next level. Rejecting marks every remaining level **Skipped** and stops the whole trail.
- A ticket awaiting approval cannot be moved into any working status, and a rejected request can only be cancelled or closed.

4.5 The IT manager

- Own **Service Desk Settings** — which modules are on, the defaults, the automation thresholds, the notification switches and audit retention.
- Own the **SLA policies**, including the holiday list and business hours. This is the single biggest lever on measured performance.
- Own the **approval matrices**. If the seeded matrices reference **Requester Manager**, they resolve through `Employee.reports_to` — if reporting lines are missing, mandatory approval levels fail with "No approver could be resolved".
- You are the second escalation rung, and the seeded 100 % rule also raises the ticket priority one step and notifies the requester.
- Live on the manager dashboard and Monthly KPI.

4.6 The CISO and the auditor

- **CISO.** Every incident of type Security, Data Breach, Data Loss, Malware, Phishing or Unauthorized Access automatically sets the security flag and alerts everyone holding the CISO role at the moment it is created. S1 and S2 incidents cannot be closed without a documented root cause. CISO is also the final approval level on privileged access, and on high or critical risk changes.

- **Auditor.** Read-only across the module, with report, export, print and e-mail rights. The Service Desk Audit Log is readable only by System Manager, IT Manager, CISO and Auditor — and it cannot be edited by anyone, or deleted by anyone but Administrator.

5. Status flows and workflows

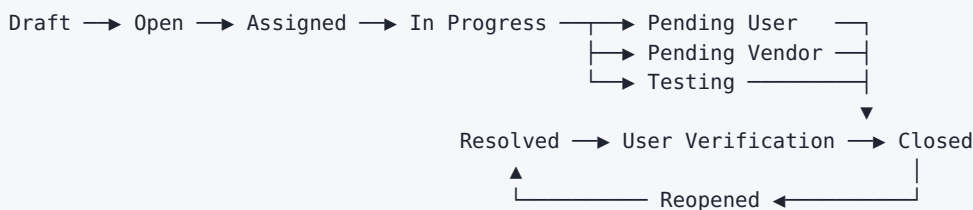
5.1 How workflow is implemented in this app

Important for administrators. This module contains **no Frappe Workflow documents** and ships none as fixtures. Do not look for entries under `/app/workflow` — there are none, and none are required.

Instead, every state machine is **enforced in the document controllers in python**. That means the rules apply identically in the desk UI, in the portal, in the REST API, in a bench console session and in scheduled jobs — there is no path around them. It also means you cannot change the flow by editing a Workflow record; changing it is a code change.

Approval, which in stock Frappe would often be modelled as a Workflow, is instead handled by the **Approval Matrix** masters plus the approval engine (section 7.3). That is what gives you reusable, condition-driven, multi-level approval across four different doctypes.

5.2 Service Ticket — the 12-state flow



Status	Meaning	SLA effect
Draft	Being written by the requester, not yet submitted to IT.	Clock runs from opened_on; the ticket is forced to Open if somebody other than the requester creates it.
Open	With IT, not yet assigned.	Running
Assigned	An engineer owns it. Set automatically the moment auto-assignment succeeds.	Running
In Progress	Actively being worked.	Running; stamps first response if not already stamped
Pending User	Waiting for information from the requester.	Paused (default policy)
Pending Vendor	Waiting for a third party.	Paused (default policy)
Testing	Fix applied, being verified by IT.	Running; stamps first response
Resolved	IT believes it is fixed. Requires a Resolution.	Resolution compliance is evaluated here
User Verification	Waiting for the requester to confirm.	Eligible for auto-close
Closed	Finished. Requires a Resolution.	Final
Reopened	The requester says it is not fixed.	Resolution clock restarts ; reopen count increments; closed date cleared

Status	Meaning	SLA effect
Cancelled	Withdrawn or invalid.	SLA set to Not Applicable ; terminal — nothing follows it

5.3 Transition matrix

Any move not listed below is rejected with "Cannot move a ticket from X to Y."

From	Allowed next statuses
Draft	Open, Cancelled
Open	Assigned, In Progress, Pending User, Pending Vendor, Resolved, Cancelled
Assigned	Open, In Progress, Pending User, Pending Vendor, Testing, Resolved, Cancelled
In Progress	Assigned, Pending User, Pending Vendor, Testing, Resolved, Cancelled
Pending User	In Progress, Assigned, Resolved, Closed, Cancelled
Pending Vendor	In Progress, Assigned, Resolved, Cancelled
Testing	In Progress, Resolved, User Verification, Cancelled
Resolved	User Verification, Closed, Reopened, In Progress
User Verification	Closed, Reopened, Resolved
Closed	Reopened
Reopened	Open, Assigned, In Progress, Resolved, Cancelled
Cancelled	none — terminal

Two further guards apply on top of the matrix:

- A ticket whose approval status is **Pending** cannot enter In Progress, Testing, Resolved, User Verification or Closed.
- A ticket whose approval status is **Rejected** can only go to Cancelled, Closed, Reopened or Resolved.

5.4 The other lifecycles

Incident

Open → Investigating → Contained → Mitigated → Resolved → Closed

- Entering **Contained** stamps containment_time; **Resolved/Closed** stamps resolution_time. These two timings drive the "Contained (h)" and "Resolved (h)" columns of the Incident Report.
- Closing requires a **Resolution**. For severity **S1 - Critical** or **S2 - High**, a **Root Cause** is also mandatory.

Problem

Open → Under Investigation → Known Error → Fix In Progress → Resolved → Closed

- Setting the status to **Known Error** sets the known-error flag; a known error without a documented workaround produces a warning.
- Resolving requires a **Root Cause**, and stamps the resolved date automatically.
- incident_count is refreshed on every save from the tickets pointing at this problem.

1. Is approval required?
 - Service Ticket → the service is flagged approval_required, OR the category/sub-category is flagged requires_approval, OR a matrix is already set on the ticket
 - Change Request → change type is not "Standard", OR cab_required is set
 - Privileged Access → always
 - Service Catalog item → whatever the matrix says
2. Which matrix? Score every active matrix whose "Applies To" matches this doctype.
 - a matrix condition that is SET and does NOT match → matrix disqualified
 - a matrix condition that is SET and DOES match → +1 point
 Conditions scored: category, department, priority, risk level, amount range.
 Highest score wins. A matrix already on the document, or on the chosen Service Catalog item, short-circuits the search.
3. Build the trail. For each level in order:
 - resolve the approver (Role · Specific User · Department Head · Team Lead · IT Manager · CISO · Requester Manager)
 - evaluate the optional python condition → false means the level is Skipped
 - no approver resolvable and the level is mandatory → the save is BLOCKED
4. Collect decisions.
 - Sequential → only the lowest pending level can act
 - Parallel → every pending level can act at once
 - Approve → level closed, move to the next; last one closed = Approved
 - Reject → whole trail stops, remaining levels marked Skipped (reason required)

Approver type	Resolves to
Role	Anybody holding the named role
Specific User	That one named user
Requester Manager	The requester's manager, found through <code>Employee.reports_to</code> → that employee's <code>user_id</code> . Falls back to the Department Head role.
Department Head	The head recorded on the document's Department, mapped back to a user
Team Lead	The <code>team_lead</code> of the ticket's assigned team
IT Manager	The IT Manager role
CISO	The CISO role

The three matrices seeded at install are:

Matrix	Applies to	Levels
Standard Service Request Approval	Service Ticket	1. Requester Manager (mandatory) 2. IT Manager (mandatory)
Standard Change Approval	Change Request	1. Team Lead (mandatory) 2. IT Manager (mandatory) 3. CISO (optional, only when <code>doc.risk in ('High', 'Critical')</code>)
Privileged Access Approval	Privileged Access Request	1. Requester Manager 2. IT Manager 3. CISO — all mandatory

6. Complete DocType listing

The module owns **36 DocTypes**: 18 transactional, master, configuration and audit documents that you open and work with, plus 18 child tables that only ever appear as grids inside a parent. This section lists every one of them, every field, and the permissions on each. Permission codes are the ones defined in section 3.3: **R** read, **W** write, **C** create, **D** delete, **Y** report, **E** export, **H** share, **P** print, **M** e-mail.

6.1 Overview of all 36

DocType	Fields	Naming	What it is
Transaction — 8 DocTypes			
Change Request	32	Series CHG- .YYYY. - .#####	A controlled modification to a live IT service, with justification, implementation plan, rollback plan, test plan, risk rating, downtime window and CAB approval.
Incident	25	Series INC- .YYYY. - .#####	An unplanned interruption or quality reduction of an IT service - an outage, a degradation, or a security event such as malware, phishing or a data breach.
Privileged Access Request	21	Series PAR- .YYYY. - .#####	A time-boxed grant of elevated access (ERP admin, SSH, firewall, VPN, cloud, database, domain admin, .
Problem	19	Series PRB- .YYYY. - .#####	The underlying cause behind one or more incidents.
Service Ticket	64	Series TKT- .YYYY. - .#####	The central record of the whole system.
Ticket Comment	6	System generated hash	A threaded note on a ticket, either a Public Note the requester can see or an Internal Note visible only to service desk roles.
Ticket Feedback	9	Format FB-{ticket}	The requester's satisfaction survey for one closed ticket: an overall rating plus response speed, resolution quality and engineer courtesy.
Work Log	11	Series WL- .YYYY. - .#####	One block of effort an engineer spent on a ticket - diagnosis, implementation, testing, communication, documentation or travel.
Master — 8 DocTypes			
Approval Matrix	11	Named after matrix_name	A reusable approval ladder - who signs off, in what order, under which conditions - for Service Tickets, Change Requests, Privileged Access Requests or Service Catalog items.
Configuration Item	17	Named after ci_name	The CMDB record for a piece of infrastructure - server, application, database, website, API, firewall, storage, network device, cloud VM or licence - with environment, criticality, owner, support team, hostname, IP, vendor, warranty and its relationships to other CIs.
Engineer Skill	6	Format SKL-{engineer}-{category}	One engineer's competence in one category: Beginner / Intermediate / Expert, plus a certification flag and an active flag.

DocType	Fields	Naming	What it is
IT Service Category	12	Named after <code>category_name</code>	The service taxonomy - a two-level tree of categories and sub-categories (Infrastructure → Server, Networking → VPN, .
Knowledge Article	19	Series KB- .YYYY. - .#####	A versioned knowledge base entry - FAQ, How-To, Video, Policy, SOP, Known Error or Troubleshooting guide - with view, helpful and not-helpful counters.
SLA Policy	11	Named after <code>policy_name</code>	Response and resolution targets per priority, the business hours they are measured in, the holiday list to skip, the statuses that pause the clock and the escalation ladder.
Service Catalog	14	Named after <code>service_name</code>	The published menu of requestable services ('New Laptop', 'VPN Access', 'Password Reset') with their category, priority, SLA policy, cost centre, estimated cost and required documents.
Service Desk Team	9	Named after <code>team_name</code>	A support group with a lead, an escalation contact, a default category and its members, each with a capacity limit and an availability flag.
Configuration — 1 DocType			
Service Desk Settings	24	Single document	The single global control panel: which ITSM modules are on, the system-wide defaults, the automation thresholds, the notification switches and the audit retention period.
Audit — 1 DocType			
Service Desk Audit Log	9	System generated hash	The append-only compliance trail.
Child table — 18 DocTypes			
Approval History	6	Child row	One row per approval level on a ticket, change or access request: the level number, the resolved approver or role, the outcome and the comment.
Approval Matrix Level	7	Child row	One rung of an approval ladder: level number, approver type (Role, Specific User, Department Head, Team Lead, IT Manager, CISO or Requester Manager), mode, mandatory flag and an optional python condition.
Assignment History	5	Child row	The audit of who a ticket was assigned to, by whom, when and why.
CI Relationship	3	Child row	A typed link from one configuration item to another - Depends On, Used By, Runs On, Hosts, Connected To, Part Of, Backed Up By.
Change Affected Service	4	Child row	A configuration item or named service a change will touch, whether it needs downtime, and the impact notes.
Notification Sent	7	Child row	The delivery log for one notification: channel, recipient, trigger event, timestamp, Sent or Failed, and the error text if it failed.
Privileged Access Scope	4	Child row	One system inside an access request: which system, which host or target, and the access level (Read / Write / Admin / Root).

DocType	Fields	Naming	What it is
Related Record	4	Child row	A free-form link to any other document with a relationship type - Related, Duplicate Of, Caused By, Blocks, Blocked By, Parent Of, Child Of.
SLA Escalation Rule	6	Child row	One rung of the escalation ladder: which clock (Response or Resolution), at what percentage of elapsed target it fires, who it goes to, whether it raises the priority and whether the requester is told.
SLA Event	3	Child row	The SLA journal of a ticket: SLA Applied, Response Met/Breached, Resolution Met/Breached, Paused, Resumed, Escalated, Reset - each with a timestamp and details.
SLA Pause Status	2	Child row	A ticket status that stops the SLA clock, with a note explaining why.
SLA Priority Rule	3	Child row	The response and resolution target for one priority, as durations.
SLA Working Day	3	Child row	One business-hours window: weekday, start time, end time.
Service Catalog Document	3	Child row	A document the requester must supply for a catalog service, and whether it is mandatory.
Service Desk Team Member	5	Child row	One member of a team: the user, their role in the team (Member / Lead / Backup), their maximum open ticket count and whether they are currently available.
Ticket Asset Link	4	Child row	An extra ERPNext Asset or configuration item involved in a ticket, beyond the single primary link on the ticket itself.
Ticket Timeline	6	Child row	The human-readable activity feed of a ticket, incident or change: time, event type, user, description and the from/to values of whatever changed.
Ticket Watcher	2	Child row	Someone who is copied on a ticket without owning it, and how much they want to hear - All Updates, Status Change only, or Resolution Only.

Transaction DocTypes

Change Request

Type	Tracks changes	Naming	Series CHG- .YYYY. - .#####
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What it is. A controlled modification to a live IT service, with justification, implementation plan, rollback plan, test plan, risk rating, downtime window and CAB approval.

How it is used. Moves Draft → Submitted to start its approval trail. High/Critical risk and Emergency changes force cab_required. It cannot be scheduled or implemented while approval is pending, and cannot be completed without a validation result. Feeds the Change Success Report.

Field	Label	Type / options	Notes
naming_series	Series	Select CHG-.YYYY-.#####	required ; default CHG- .YYYY. - .#####
title	Title	Data	required
change_type	Change Type	Select Standard Normal Emergency	required ; default Normal
risk	Risk	Select Low Medium High Critical	required ; default Medium
status	Status	Select Draft Submitted Risk Review CAB Approval Scheduled Implementation Validation Completed Closed Rolled Back Rejected Cancelled	required ; default Draft
priority	Priority	Select Low Medium High Critical	default Medium
category	Category	Link → IT Service Category	
requested_by	Requested By	Link → User	required
change_owner	Change Owner	Link → User	
Justification & Plans			
business_justification	Business Justification	Text	required
implementation_plan	Implementation Plan	Text	required
rollback_plan	Rollback Plan	Text	required
testing_plan	Testing Plan	Text	
Impact & Window			
downtime_required	Downtime Required	Check	default 0
downtime_minutes	Downtime (minutes)	Int	shown when downtime_required

Field	Label	Type / options	Notes
planned_start	Planned Start	Datetime	
planned_end	Planned End	Datetime	
actual_start	Actual Start	Datetime	read-only
actual_end	Actual End	Datetime	read-only
affected_services	Affected Services	Table → Change Affected Service	
Approval			
cab_required	CAB Approval Required	Check	default 0
cab_date	CAB Meeting Date	Date	shown when cab_required
approval_matrix	Approval Matrix	Link → Approval Matrix	
approval_status	Approval Status	Select Not Required Pending Approved Rejected	read-only; default Not Required
current_approval_level	Current Approval Level	Int	read-only; default 0
approvals	Approvals	Table → Approval History	read-only
Implementation Result			
implementation_notes	Implementation Notes	Text	
validation_result	Validation Result	Select (blank) Successful Successful with Issues Failed Rolled Back	
rollback_reason	Rollback Reason	Text	shown when eval:doc.validation_result=='Rolled Back' doc.status=='Rolled Back'
Related & Timeline			
related_records	Related Records	Table → Related Record	
timeline	Timeline	Table → Ticket Timeline	read-only
ticket	Originating Ticket	Link → Service Ticket	read-only

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM

Role	Rights
Service Desk Executive	RYPM
Department Head	RYPM
CISO	RYPM
Auditor	RYEPM

Incident

Type	Tracks changes	Naming	Series <code>INC-.YYYY.-.#####</code>
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What it is. An unplanned interruption or quality reduction of an IT service - an outage, a degradation, or a security event such as malware, phishing or a data breach.

How it is used. Raised standalone, from a ticket via the **Create → Incident** button, or through `api.itsm.create_incident`. Security types automatically set `is_security_incident` and alert everyone holding the CISO role. S1/S2 incidents cannot be closed without a root cause.

Field	Label	Type / options	Notes
<code>naming_series</code>	Series	Select INC-.YYYY.-.#####	required ; default <code>INC-.YYYY.-.#####</code>
<code>title</code>	Title	Data	required
<code>incident_type</code>	Incident Type	Select Outage Degradation Security Data Breach Data Loss Malware Phishing Unauthorized Access Hardware Failure Other	required
<code>severity</code>	Severity	Select S1 - Critical S2 - High S3 - Medium S4 - Low	required ; default <code>S3 - Medium</code>
<code>status</code>	Status	Select Open Investigating Contained Mitigated Resolved Closed	required ; default <code>Open</code>
<code>is_security_incident</code>	Security Incident	Check	default <code>0</code>
<code>affected_service</code>	Affected Service (CI)	Link → Configuration Item	
<code>affected_users</code>	Affected Users	Int	
<code>reported_by</code>	Reported By	Link → User	
<code>incident_manager</code>	Incident Manager	Link → User	
<code>problem</code>	Problem	Link → Problem	
Timeline			
<code>detection_time</code>	Detected At	Datetime	
<code>reported_time</code>	Reported At	Datetime	
<code>containment_time</code>	Contained At	Datetime	
<code>resolution_time</code>	Resolved At	Datetime	
Impact			
<code>business_impact</code>	Business Impact	Text	
Response			

Field	Label	Type / options	Notes
containment_actions	Containment Actions	Text	
resolution	Resolution	Text	
Root Cause Analysis			
root_cause	Root Cause	Text	
lessons_learned	Lessons Learned	Text	
capa	Corrective & Preventive Action (CAPA)	Text	
rca_document	RCA Document	Attach	
Related & Timeline			
related_records	Related Tickets	Table → Related Record	
timeline	Timeline	Table → Ticket Timeline	read-only
ticket	Originating Ticket	Link → Service Ticket	read-only

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
CISO	RWCYHPM
Service Desk Executive	RYPM
Auditor	RYEPM

Privileged Access Request

Type	Tracks changes	Naming	Series PAR-.YYYY.-.#####
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What it is. A time-boxed grant of elevated access (ERP admin, SSH, firewall, VPN, cloud, database, domain admin, ...) with an explicit scope, justification and expiry.

How it is used. Always requires approval. Once every level approves, the request is granted and stamped; the hourly job revokes or expires it the moment valid_till passes, and the daily job warns one day beforehand. Every grant and revocation is written to the audit log.

Field	Label	Type / options	Notes
naming_series	Series	Select PAR-.YYYY.-.#####	required ; default PAR-.YYYY.-.#####
requester	Requester	Link → User	required
requested_for	Requested For	Link → User	
request_type	Primary Access Type	Select ERP Admin SSH Firewall VPN Azure AWS Database GitHub Office365 Domain Admin Other	required

Field	Label	Type / options	Notes
status	Status	Select Draft Pending Approval Approved Active Expired Revoked Rejected Cancelled	required ; default Draft
ticket	Related Ticket	Link → Service Ticket	
valid_from	Valid From	Datetime	required
valid_till	Valid Till	Datetime	required
auto_revoke	Auto Revoke On Expiry	Check	default 1
Access Scope			
access_scope	Systems	Table → Privileged Access Scope	required
Justification			
business_justification	Business Justification	Text	required
Approval			
approval_matrix	Approval Matrix	Link → Approval Matrix	
approval_status	Approval Status	Select Not Required Pending Approved Rejected	read-only; default Pending
current_approval_level	Current Level	Int	read-only; default 0
approvals	Approvals	Table → Approval History	read-only
Grant & Revocation			
granted_by	Granted By	Link → User	read-only
granted_on	Granted On	Datetime	read-only
revoked_by	Revoked By	Link → User	read-only
revoked_on	Revoked On	Datetime	read-only
revocation_reason	Revocation Reason	Small Text	
Audit			
audit_notes	Audit Notes	Text	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
CISO	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RYPM
Employee	RWCYPM
Department Head	RYPM
Auditor	RYEPM

Problem

Type	Tracks changes	Naming	Series PRB- .YYYY. - .#####
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What it is. The underlying cause behind one or more incidents. Holds trend analysis, root cause, a workaround and the permanent fix.

How it is used. Created by hand, from a ticket, or automatically by the daily job when more than N incidents (default 5) hit the same category within 30 days. Cannot be resolved without a root cause.

Field	Label	Type / options	Notes
<code>naming_series</code>	Series	Select PRB-.YYYY.-.#####	required ; default PRB-.YYYY.-.#####
<code>title</code>	Title	Data	required
<code>status</code>	Status	Select Open Under Investigation Known Error Fix In Progress Resolved Closed	required ; default Open
<code>priority</code>	Priority	Select Low Medium High Critical	default Medium
<code>category</code>	Category	Link → IT Service Category	
<code>assigned_to</code>	Assigned To	Link → User	
<code>identified_on</code>	Identified On	Date	
<code>resolved_on</code>	Resolved On	Date	read-only
<code>is_known_error</code>	Known Error	Check	default 0
Description			
<code>description</code>	Description	Text	
Analysis			
<code>incident_count</code>	Linked Incident Count	Int	read-only; default 0
<code>trend_analysis</code>	Trend Analysis	Text	
<code>root_cause</code>	Root Cause	Text	
<code>workaround</code>	Workaround	Text	
<code>permanent_fix</code>	Permanent Fix	Text	
Closure			
<code>knowledge_article</code>	Knowledge Article	Link → Knowledge Article	
<code>closure_notes</code>	Closure Notes	Text	
Related Records			
<code>related_records</code>	Related Incidents & Tickets	Table → Related Record	
<code>change_request</code>	Change Request	Link → Change Request	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM

Role	Rights
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RYPM
CISO	RYPM
Auditor	RYEPM

Service Ticket

Type	Tracks changes	Naming	Series	TKT-.YYYY.-.#####
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What it is. The central record of the whole system. Every request, incident, query or complaint that reaches IT becomes one Service Ticket, whatever channel it arrived on (portal, e-mail, phone, chat, walk-in, API).

How it is used. Created from the `/support` portal, the desk form, or `api.ticket.create`. On every save its controller derives priority, validates the status transition, resolves approval, routes the ticket to a team and engineer, applies the SLA and writes the timeline. It is the reference doctype for 9 of the 12 reports and for most number cards.

Field	Label	Type / options	Notes
<code>naming_series</code>	Series	Select TKT-.YYYY.-.#####	required ; default <code>TKT-.YYYY.-.#####</code>
<code>subject</code>	Subject	Data	required
<code>ticket_type</code>	Ticket Type	Select Incident Service Request Change Request Problem Query	required ; default <code>Service Request</code>
<code>status</code>	Status	Select Draft Open Assigned In Progress Pending User Pending Vendor Testing Resolved User Verification Closed Reopened Cancelled	required ; default <code>Open</code>
<code>priority</code>	Priority	Select Low Medium High Critical	required ; default <code>Medium</code>
<code>service</code>	Service (Catalog)	Link → Service Catalog	
<code>category</code>	Category	Link → IT Service Category	required
<code>sub_category</code>	Sub Category	Link → IT Service Category	
<code>source</code>	Source	Select Portal Email Phone Chat Walk-in API Auto	default <code>Portal</code>
<code>department</code>	Department	Link → Department	
Requester			
<code>requester</code>	Requester	Link → User	required
<code>requester_name</code>	Requester Name	Data	read-only; fetched from <code>requester.full_name</code>

Field	Label	Type / options	Notes
requested_for	Requested For	Link → User	
contact_number	Contact Number	Data	
requester_employee	Employee	Link → Employee	
Assignment			
assigned_team	Assigned Team	Link → Service Desk Team	
assigned_engineer	Assigned Engineer	Link → User	
engineer_name	Engineer Name	Data	read-only; fetched from <code>assigned_engineer.full_name</code>
impact	Impact	Select Low Medium High	default Medium
urgency	Urgency	Select Low Medium High	default Medium
severity	Severity	Select (blank) S1 - Critical S2 - High S3 - Medium S4 - Low	
Description			
short_description	Short Description	Small Text	
description	Detailed Description	Text Editor	
SLA			
sla_policy	SLA Policy	Link → SLA Policy	read-only
sla_status	SLA Status	Select Ongoing Paused Fulfilled Failed Response Overdue Resolution Overdue Not Applicable	read-only; default Ongoing
response_by	Respond By	Datetime	read-only
resolution_by	Resolve By	Datetime	read-only
first_responded_on	First Responded On	Datetime	read-only
response_sla_met	Response SLA Met	Check	read-only; default 0
resolution_sla_met	Resolution SLA Met	Check	read-only; default 0
total_hold_time	Total Hold Time	Duration	read-only; default 0
hold_started_on	Hold Started On	Datetime	read-only
is_escalated	Escalated	Check	read-only; default 0
escalation_level	Escalation Level	Int	read-only; default 0
Dates			
opened_on	Opened On	Datetime	read-only
resolved_on	Resolved On	Datetime	read-only
closed_on	Closed On	Datetime	read-only
reopened_on	Last Reopened On	Datetime	read-only
reopen_count	Reopen Count	Int	read-only; default 0

Field	Label	Type / options	Notes
Approval			
approval_status	Approval Status	Select Not Required Pending Approved Rejected	read-only; default Not Required
approval_matrix	Approval Matrix	Link → Approval Matrix	read-only
current_approval_level	Current Level	Int	read-only; default 0
approvals	Approvals	Table → Approval History	read-only
Resolution			
root_cause	Root Cause	Text	
resolution	Resolution	Text Editor	
resolution_category	Resolution Category	Select (blank) Fixed Workaround Provided Configuration Change User Education No Fault Found Duplicate Cancelled	
knowledge_article	Knowledge Article Used	Link → Knowledge Article	
Feedback			
feedback_rating	Rating	Rating	read-only
feedback_comments	Feedback Comments	Small Text	read-only
Linked Records			
linked_asset	Asset	Link → Asset	
configuration_item	Configuration Item	Link → Configuration Item	
linked_project	Project	Link → Project	
linked_incident	Incident	Link → Incident	
linked_problem	Problem	Link → Problem	
linked_change	Change Request	Link → Change Request	
Assets & Related			
asset_links	Additional Assets	Table → Ticket Asset Link	
related_records	Related Records	Table → Related Record	
watchers	Watchers	Table → Ticket Watcher	
History			
timeline	Timeline	Table → Ticket Timeline	read-only
sla_events	SLA Events	Table → SLA Event	read-only
assignment_history	Assignment History	Table → Assignment History	read-only
notifications_sent	Notifications Sent	Table → Notification Sent	read-only
Metrics			

Field	Label	Type / options	Notes
total_worked_hours	Total Worked Hours	Float	read-only; default 0
tags	Tags	Data	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RWCYHPM
Employee	RWCYHPM
Department Head	RYPM
CISO	RYPM
Auditor	RYEPM

Ticket Comment

Type	Tracks changes	Naming	System generated hash
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What it is. A threaded note on a ticket, either a **Public Note** the requester can see or an **Internal Note** visible only to service desk roles.

How it is used. Inserting a comment appends a timeline row, notifies the counterparty (public notes only) and, when written by an engineer, stamps the SLA first response. The REST `comments` endpoint hides internal notes from users without a service desk role.

Field	Label	Type / options	Notes
<code>ticket</code>	Ticket	Link → Service Ticket	required
<code>comment_by</code>	Comment By	Link → User	required
<code>comment_type</code>	Type	Select Public Note Internal Note	required ; default <code>Public Note</code>
<code>commented_on</code>	Commented On	Datetime	read-only
<code>content</code>	Comment	Text Editor	required
<code>attachment</code>	Attachment	Attach	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RWCYHPM
Employee	RWCYPM
Auditor	RYEPM

Ticket Feedback

Type	Tracks changes	Naming	Format <code>FB-{ticket}</code>
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What it is. The requester's satisfaction survey for one closed ticket: an overall rating plus response speed, resolution quality and engineer courtesy.

How it is used. One per ticket (the ticket field is unique). Can only be submitted once the ticket is Resolved, User Verification or Closed. On save it pushes the rating back onto the ticket, which is what Engineer Performance, Monthly KPI and the executive dashboard read.

Field	Label	Type / options	Notes
<code>ticket</code>	Ticket	Link → Service Ticket	required ; unique
<code>requester</code>	Requester	Link → User	read-only
<code>submitted_on</code>	Submitted On	Datetime	read-only
<code>rating</code>	Overall Rating	Rating	required
<code>would_recommend</code>	Would Recommend	Check	default <code>1</code>
Detailed Ratings			
<code>response_speed</code>	Response Speed	Rating	
<code>resolution_quality</code>	Resolution Quality	Rating	
<code>engineer_courtesy</code>	Engineer Courtesy	Rating	
<code>comments</code>	Comments	Small Text	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RYPM
Service Desk Engineer	RYPM
Service Desk Executive	RYPM
Employee	RWCYPM
Auditor	RYEPM

Work Log

Type	Tracks changes	Naming	Series WL-.YYYY.-.#####
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What it is. One block of effort an engineer spent on a ticket - diagnosis, implementation, testing, communication, documentation or travel.

How it is used. Hours are computed from start/end time. On save it rolls the total up into the ticket's total_worked_hours, appends a timeline row and - if this is the first engineer activity - stamps the SLA first response. Drives the 'Logged Hours' column of Engineer Performance.

Field	Label	Type / options	Notes
naming_series	Series	Select WL-.YYYY.-.#####	required ; default WL-.YYYY.-.#####
ticket	Ticket	Link → Service Ticket	required
engineer	Engineer	Link → User	required
activity_type	Activity	Select Diagnosis Implementation Testing Communication Documentation Travel Other	default Diagnosis
start_time	Start Time	Datetime	required
end_time	End Time	Datetime	
hours	Hours	Float	read-only
is_billable	Billable	Check	default 0
is_internal	Internal Only	Check	default 0
description	Description	Text	required
attachment	Attachment	Attach	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RWCYHPM
Auditor	RYEPM

Master DocTypes

Approval Matrix

Type	Tracks changes	Naming	Named after <code>matrix_name</code>
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What it is. A reusable approval ladder - who signs off, in what order, under which conditions - for Service Tickets, Change Requests, Privileged Access Requests or Service Catalog items.

How it is used. The engine scores every active matrix against the document's category, department, priority, risk and amount; the most specific match wins. Levels can be Sequential or Parallel and may carry a python condition. 3 matrices are seeded at install.

Field	Label	Type / options	Notes
<code>matrix_name</code>	Matrix Name	Data	required ; unique
<code>applies_to</code>	Applies To	Select Service Ticket Change Request Privileged Access Request Service Catalog	required ; default <code>Service Ticket</code>
<code>is_active</code>	Active	Check	default <code>1</code>
<code>approval_mode</code>	Approval Mode	Select Sequential Parallel	default <code>Sequential</code>
<code>department</code>	Department	Link → Department	
Match Conditions			
<code>priority</code>	Priority	Select (blank) Low Medium High Critical	
<code>risk_level</code>	Risk Level	Select (blank) Low Medium High Critical	
<code>min_amount</code>	Minimum Amount	Currency	
<code>max_amount</code>	Maximum Amount	Currency	
Approval Levels			
<code>levels</code>	Levels	Table → Approval Matrix Level	required
<code>category</code>	Category	Link → IT Service Category	

Permissions

Role	Rights
System Manager	RWCDYEHMPM
IT Manager	RWCDYEHMPM
Service Desk Team Lead	RYPM
Department Head	RYPM
CISO	RYPM
Auditor	RYEPM

Configuration Item

Type	Tracks changes	Naming	Named after <code>ci_name</code>
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What it is. The CMDB record for a piece of infrastructure - server, application, database, website, API, firewall, storage, network device, cloud VM or licence - with environment, criticality, owner, support team, hostname, IP, vendor, warranty and its relationships to other CIs.

How it is used. Linked from tickets, incidents and changes so impact is visible. Self-references and duplicate relationships are rejected, and a CI cannot be retired while open tickets still point at it. Optionally mirrors an ERPNext Asset.

Field	Label	Type / options	Notes
ci_name	Name	Data	required ; unique
ci_type	Type	Select Server Application Database Website API Firewall Storage Network Device Cloud VM License Other	required
environment	Environment	Select Production UAT Development DR	required ; default Production
status	Status	Select Active Inactive Under Maintenance Retired	default Active
criticality	Criticality	Select Low Medium High Critical	default Medium
owner_user	Owner	Link → User	
support_team	Support Team	Link → Service Desk Team	
linked_asset	ERP Asset	Link → Asset	
Technical			
hostname	Hostname	Data	
ip_address	IP Address	Data	
operating_system	Operating System	Data	
location	Location	Data	
vendor	Vendor	Link → Supplier	
go_live_date	Go Live Date	Date	
warranty_expiry	Warranty / AMC Expiry	Date	
description	Description	Text	
relationships	Relationships	Table → CI Relationship	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RYPM
CISO	RYPM
Auditor	RYEPM

Engineer Skill

Type	Tracks changes	Naming	Format SKL-{engineer}-{category}
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What it is. One engineer's competence in one category: Beginner / Intermediate / Expert, plus a certification flag and an active flag.

How it is used. Only consulted when a team uses **Skill Based** assignment. Ranking is: exact sub-category match first, then Expert > Intermediate > Beginner, then certified over uncertified, then lowest current load. One row per engineer per category is enforced.

Field	Label	Type / options	Notes
engineer	Engineer	Link → User	required
category	Category	Link → IT Service Category	required
skill_level	Skill Level	Select Beginner Intermediate Expert	required ; default Intermediate
is_certified	Certified	Check	default 0
is_active	Active	Check	default 1
notes	Notes	Small Text	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Executive	RYPM
Service Desk Engineer	RYPM
Auditor	RYEPM

IT Service Category

Type	Tracks changes	Naming	Named after category_name
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What it is. The service taxonomy - a two-level tree of categories and sub-categories (Infrastructure → Server, Networking → VPN, ...). Named IT Service Category because plain 'Service Category' is already taken by the Turacoz Management System module.

How it is used. Everything routes off this: default priority/impact/urgency, default team, default engineer, default SLA policy and whether the category needs approval. Loops in the hierarchy are rejected, and a category with children cannot be deleted. 60 rows are seeded at install.

Field	Label	Type / options	Notes
category_name	Category Name	Data	required ; unique
parent_service_category	Parent Category	Link → IT Service Category	
is_group	Is Group	Check	default 0
is_active	Active	Check	default 1
default_priority	Default Priority	Select (blank) Low Medium High Critical	default Medium

Field	Label	Type / options	Notes
default_impact	Default Impact	Select (blank) Low Medium High	
default_urgency	Default Urgency	Select (blank) Low Medium High	
Routing Defaults			
default_team	Default Team	Link → Service Desk Team	
default_engineer	Default Engineer	Link → User	
default_sla_policy	Default SLA Policy	Link → SLA Policy	
requires_approval	Requires Approval	Check	default 0
description	Description	Small Text	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Executive	RYPM
Service Desk Engineer	RYPM
Employee	RYPM
Auditor	RYEPM

Knowledge Article

Type	Tracks changes	Naming	Series KB- .YYYY. - .#####
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What it is. A versioned knowledge base entry - FAQ, How-To, Video, Policy, SOP, Known Error or Troubleshooting guide - with view, helpful and not-helpful counters.

How it is used. Suggested to engineers and portal users by keyword overlap with the ticket text. Publishing stamps the approver and bumps the version; editing a published article bumps it again. **Create Knowledge Article** on a resolved ticket drafts one from the resolution.

Field	Label	Type / options	Notes
naming_series	Series	Select KB-.YYYY.-.#####	required ; default KB- .YYYY. - .#####
title	Title	Data	required
article_type	Type	Select FAQ How-To Video Policy SOP Known Error Troubleshooting	required ; default How-To
status	Status	Select Draft Under Review Published Archived	default Draft
category	Category	Link → IT Service Category	

Field	Label	Type / options	Notes
sub_category	Sub Category	Link → IT Service Category	
is_public	Visible On Portal	Check	default 0
version	Version	Int	read-only; default 1
Content			
content	Content	Text Editor	
video_url	Video URL	Data	shown when eval:doc.article_type=='Video'
attachment	Attachment	Attach	
Ownership & Approval			
author	Author	Link → User	
approved_by	Approved By	Link → User	read-only
approved_on	Approved On	Datetime	read-only
view_count	Views	Int	read-only; default 0
helpful_count	Marked Helpful	Int	read-only; default 0
not_helpful_count	Marked Not Helpful	Int	read-only; default 0
keywords	Keywords	Small Text	
Related Records			
related_records	Related Records	Table → Related Record	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Engineer	RWCYHPM
Service Desk Executive	RWCYHPM
Employee	RYPM
Auditor	RYEPM

SLA Policy

Type	Tracks changes	Naming	Named after policy_name
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What it is. Response and resolution targets per priority, the business hours they are measured in, the holiday list to skip, the statuses that pause the clock and the escalation ladder.

How it is used. Resolved onto a ticket in order: service → sub-category → category → the policy's own apply_to_category rule → Service Desk Settings default → the policy flagged is_default. Only one policy can be default - saving one clears the flag on all others.

Field	Label	Type / options	Notes
policy_name	Policy Name	Data	required ; unique
is_default	Default Policy	Check	default 0
is_active	Active	Check	default 1
holiday_list	Holiday List	Link → Holiday List	
apply_24x7	24x7 (ignore working hours)	Check	default 0
Response & Resolution Targets			
priority_rules	Priority Rules	Table → SLA Priority Rule	required
Business Hours			
working_days	Working Days	Table → SLA Working Day	
Pause Conditions			
pause_statuses	Pause On These Statuses	Table → SLA Pause Status	
Escalation			
escalation_rules	Escalation Rules	Table → SLA Escalation Rule	
description	Description	Small Text	
apply_to_category	Apply To Category	Link → IT Service Category	

Permissions

Role	Rights
System Manager	RWCDYEHMPM
IT Manager	RWCDYEHMPM
Service Desk Team Lead	RYPM
Service Desk Executive	RYPM
Service Desk Engineer	RYPM
Auditor	RYEPM

Service Catalog

Type	Tracks changes	Naming	Named after service_name
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What it is. The published menu of requestable services ('New Laptop', 'VPN Access', 'Password Reset') with their category, priority, SLA policy, cost centre, estimated cost and required documents.

How it is used. Shown on the /support portal. Picking a service on a ticket pre-fills category, sub-category, priority and department, and - if the service is marked approval_required - selects the approval matrix. 16 services are seeded at install.

Field	Label	Type / options	Notes
service_name	Service Name	Data	required ; unique
category	Category	Link → IT Service Category	required
sub_category	Sub Category	Link → IT Service Category	

Field	Label	Type / options	Notes
is_active	Active	Check	default 1
default_priority	Default Priority	Select (blank) Low Medium High Critical	default Medium
sla_policy	SLA Policy	Link → SLA Policy	
department	Department	Link → Department	
cost_center	Cost Center	Link → Cost Center	
estimated_cost	Estimated Cost	Currency	
Description			
description	Description	Text Editor	
image	Icon	Attach Image	
Approval			
approval_required	Approval Required	Check	default 0
approval_matrix	Approval Matrix	Link → Approval Matrix	shown when approval_required
Required Documents			
required_documents	Required Documents	Table → Service Catalog Document	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Executive	RYPM
Service Desk Engineer	RYPM
Employee	RYPM
Auditor	RYEPM

Service Desk Team

Type	Tracks changes	Naming	Named after <code>team_name</code>
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What it is. A support group with a lead, an escalation contact, a default category and its members, each with a capacity limit and an availability flag.

How it is used. The team's `assignment_method` decides how the engine picks an engineer: Round Robin, Load Balancing (respects Max Open Tickets), Skill Based (via Engineer Skill) or Manual. The team lead is always forced into the member list.

Field	Label	Type / options	Notes
<code>team_name</code>	Team Name	Data	required ; unique
<code>team_lead</code>	Team Lead	Link → User	required
<code>email</code>	Team Email	Data	
<code>assignment_method</code>	Assignment Method	Select Round Robin Load Balancing Skill Based Manual	default <code>Round Robin</code>
<code>is_active</code>	Active	Check	default <code>1</code>
<code>escalation_user</code>	Escalation Contact	Link → User	
Members			
<code>members</code>	Members	Table → Service Desk Team Member	
<code>description</code>	Description	Small Text	
<code>default_category</code>	Default Category	Link → IT Service Category	

Permissions

Role	Rights
System Manager	RWCDYEHPM
IT Manager	RWCDYEHPM
Service Desk Team Lead	RWCYHPM
Service Desk Executive	RYPM
Service Desk Engineer	RYPM
Auditor	RYEPM

Configuration DocTypes

Service Desk Settings

Type	Single	Naming	Single document
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What it is. The single global control panel: which ITSM modules are on, the system-wide defaults, the automation thresholds, the notification switches and the audit retention period.

How it is used. Read by every engine through `utils.setting()`. Turning a module off makes the matching REST endpoints refuse to create records and stops the related scheduled jobs. Saving clears the document cache so changes take effect immediately.

Field	Label	Type / options	Notes
Enabled Modules			
<code>enable_incident_management</code>	Incident Management	Check	default <code>1</code>
<code>enable_problem_management</code>	Problem Management	Check	default <code>1</code>
<code>enable_change_management</code>	Change Management	Check	default <code>1</code>
<code>enable_privileged_access</code>	Privileged Access	Check	default <code>1</code>
<code>enable_knowledge_base</code>	Knowledge Base	Check	default <code>1</code>
<code>enable_cmdb</code>	CMDB	Check	default <code>1</code>
Defaults			
<code>default_sla_policy</code>	Default SLA Policy	Link → SLA Policy	
<code>default_team</code>	Default Team	Link → Service Desk Team	
<code>default_priority</code>	Default Priority	Select Low Medium High Critical	default <code>Medium</code>
<code>auto_assign</code>	Auto Assign New Tickets	Check	default <code>1</code>
<code>auto_apply_sla</code>	Auto Apply SLA	Check	default <code>1</code>
<code>derive_priority_from_impact_urgency</code>	Derive Priority From Impact & Urgency	Check	default <code>1</code>
Automation			
<code>auto_close_enabled</code>	Auto Close Resolved Tickets	Check	default <code>1</code>
<code>auto_close_after_days</code>	Auto Close After (days)	Int	default <code>3</code> ; shown when <code>auto_close_enabled</code>
<code>reminder_before_breach_percent</code>	Send Breach Reminder At (% elapsed)	Int	default <code>75</code>
<code>auto_create_problem_threshold</code>	Auto Create Problem After N Similar Incidents	Int	default <code>5</code>
<code>suggest_knowledge_articles</code>	Suggest Knowledge Articles	Check	default <code>1</code>

Field	Label	Type / options	Notes
satisfaction_survey	Send Satisfaction Survey On Close	Check	default 1
Notifications			
send_email_notifications	Email Notifications	Check	default 1
send_system_notifications	System Notifications	Check	default 1
notify_on_sla_breach	Notify On SLA Breach	Check	default 1
notify_watchers	Notify Watchers	Check	default 1
Audit			
enable_audit_log	Enable Audit Log	Check	default 1
audit_retention_days	Audit Retention (days)	Int	default 2555

Permissions

Role	Rights
System Manager	RWCDHPM
IT Manager	RWCDHPM
Auditor	RPM

Audit DocTypes

Service Desk Audit Log

Type	Standard	Naming	System generated hash
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What it is. The append-only compliance trail. One row per create, modify, status change, assignment, approval, rejection, access grant/revoke or delete on 13 tracked doctypes.

How it is used. Written by the `doc_events` hooks with the acting user, their IP, their user agent and a JSON diff of every changed field. The controller refuses edits outright and refuses deletes from anyone but Administrator. The daily job purges rows older than `audit_retention_days` (default 2555 = 7 years, for ISO 27001 / SOC 2).

Field	Label	Type / options	Notes
timestamp	Timestamp	Datetime	read-only
reference_doctype	Type	Link → DocType	read-only
reference_name	Record	Dynamic Link → reference_doctype	read-only
action	Action	Select Created Modified Assigned Status Changed Approved Rejected Comment Attachment Deleted Exported Viewed Escalated SLA Event Access Granted Access Revoked API Call	read-only
user	User	Link → User	read-only
user_full_name	User Name	Data	read-only
ip_address	IP Address	Data	read-only
browser	User Agent	Small Text	read-only
details	Details	Code	read-only

Permissions

Role	Rights
System Manager	RYEPM
IT Manager	RYEPM
CISO	RYEPM
Auditor	RYEPM

Child table DocTypes

Approval History

Type	Child table	Naming	Child row
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What it is. One row per approval level on a ticket, change or access request: the level number, the resolved approver or role, the outcome and the comment.

How it is used. Built by `approval.start()` and updated by approve/reject. Read-only on the form. Queried directly by `get_pending_approvals()` to build every user's approval inbox.

Field	Label	Type / options	Notes
<code>approval_level</code>	Level	Int	
<code>approver</code>	Approver	Link → User	
<code>approver_role</code>	Role	Link → Role	
<code>status</code>	Status	Select Pending Approved Rejected Skipped	default Pending
<code>action_on</code>	Action On	Datetime	
<code>comments</code>	Comments	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Approval Matrix Level

Type Child table **Naming** Child row

What it is. One rung of an approval ladder: level number, approver type (Role, Specific User, Department Head, Team Lead, IT Manager, CISO or Requester Manager), mode, mandatory flag and an optional python condition.

How it is used. Levels are sorted by number and resolved to a concrete user or role at the moment approval starts. A level whose condition evaluates false is recorded as **Skipped**. A mandatory level with no resolvable approver blocks the save with an error.

Field	Label	Type / options	Notes
level	Level	Int	required
approver_type	Approver Type	Select Role Specific User Department Head Team Lead IT Manager CISO Requester Manager	required ; default Role
approver_role	Role	Link → Role	shown when <code>eval:doc.approver_type=='Role'</code>
approver_user	User	Link → User	shown when <code>eval:doc.approver_type=='Specific User'</code>
approval_mode	Mode	Select Sequential Parallel	default Sequential
is_mandatory	Mandatory	Check	default 1
condition	Condition	Code	

Permissions

Child table — permissions are inherited from the parent document.

Assignment History

Type Child table **Naming** Child row

What it is. The audit of who a ticket was assigned to, by whom, when and why.

How it is used. Appended on every automatic and manual assignment. Read-only on the form.

Field	Label	Type / options	Notes
assigned_to	Assigned To	Link → User	
team	Team	Link → Service Desk Team	
assigned_by	Assigned By	Link → User	
assigned_on	Assigned On	Datetime	
reason	Reason	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

CI Relationship

Type	Child table	Naming	Child row
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What it is. A typed link from one configuration item to another - Depends On, Used By, Runs On, Hosts, Connected To, Part Of, Backed Up By.

How it is used. Used on Configuration Item to build the dependency graph. Self-references and duplicates are rejected on save.

Field	Label	Type / options	Notes
relationship_type	Relationship	Select Depends On Used By Runs On Hosts Connected To Part Of Backed Up By	required
configuration_item	Configuration Item	Link → Configuration Item	required
notes	Notes	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Change Affected Service

Type	Child table	Naming	Child row
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What it is. A configuration item or named service a change will touch, whether it needs downtime, and the impact notes.

How it is used. Used on Change Request to document blast radius for the CAB.

Field	Label	Type / options	Notes
configuration_item	Configuration Item	Link → Configuration Item	
service_name	Service	Data	
downtime_required	Downtime	Check	default 0
impact_notes	Impact	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Notification Sent

Type	Child table	Naming	Child row
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What it is. The delivery log for one notification: channel, recipient, trigger event, timestamp, Sent or Failed, and the error text if it failed.

How it is used. Written by the notification engine on every send. This is where you look when a user says they never got the e-mail. The channel field already carries SMS, Teams and WhatsApp; only Email and System are wired up today.

Field	Label	Type / options	Notes
channel	Channel	Select Email System SMS Teams WhatsApp	default Email
recipient	Recipient	Data	
trigger_event	Trigger	Data	
sent_on	Sent On	Datetime	
status	Status	Select Sent Failed	default Sent
subject	Subject	Data	
error	Error	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Privileged Access Scope

Type	Child table	Naming	Child row
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What it is. One system inside an access request: which system, which host or target, and the access level (Read / Write / Admin / Root).

How it is used. At least one row is mandatory on a Privileged Access Request. The list of systems is copied into the audit log entry when access is granted.

Field	Label	Type / options	Notes
system	System	Select ERP Admin SSH Firewall VPN Azure AWS Database GitHub Office365 Domain Admin Other	required
target	Target / Host	Data	
access_level	Access Level	Select Read Write Admin Root	default Read
notes	Notes	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Related Record

Type	Child table	Naming	Child row
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What it is. A free-form link to any other document with a relationship type - Related, Duplicate Of, Caused By, Blocks, Blocked By, Parent Of, Child Of.

How it is used. Present on Service Ticket, Incident, Problem, Change Request and Knowledge Article. Uses a Dynamic Link so it can point at any doctype.

Field	Label	Type / options	Notes
reference_doctype	Type	Link → DocType	required
reference_name	Record	Dynamic Link → reference_doctype	required
relationship	Relationship	Select Related Duplicate Of Caused By Blocks Blocked By Parent Of Child Of	default Related
notes	Notes	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

SLA Escalation Rule

Type	Child table	Naming	Child row
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What it is. One rung of the escalation ladder: which clock (Response or Resolution), at what percentage of elapsed target it fires, who it goes to, whether it raises the priority and whether the requester is told.

How it is used. Evaluated every 15 minutes. A `rule:<n>` marker written into the SLA Events table stops a rule from firing twice on the same ticket.

Field	Label	Type / options	Notes
escalation_type	Applies To	Select Response Resolution	required ; default Resolution
after_percent	Trigger At % Elapsed	Percent	required ; default 75
escalate_to_role	Escalate To Role	Link → Role	
escalate_to_user	Escalate To User	Link → User	
increase_priority	Raise Priority	Check	default 0
notify_requester	Notify Requester	Check	default 0

Permissions

Child table — permissions are inherited from the parent document.

SLA Event

Type	Child table	Naming	Child row
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What it is. The SLA journal of a ticket: SLA Applied, Response Met/Breached, Resolution Met/Breached, Paused, Resumed, Escalated, Reset - each with a timestamp and details.

How it is used. Read-only. It is both the human explanation of what the SLA engine did and the idempotency store that prevents repeat escalations and repeat reminders.

Field	Label	Type / options	Notes
event	Event	Select SLA Applied Response Met Response Breached Resolution Met Resolution Breached Paused Resumed Escalated Reset	
event_time	Time	Datetime	
details	Details	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

SLA Pause Status

Type	Child table	Naming	Child row
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What it is. A ticket status that stops the SLA clock, with a note explaining why.

How it is used. Seeded with **Pending User** and **Pending Vendor**. Entering one of these statuses starts a hold; leaving it adds the held working time back onto both deadlines.

Field	Label	Type / options	Notes
status	Status	Select Draft Open Assigned In Progress Pending User Pending Vendor Testing Resolved User Verification Closed Reopened Cancelled	required
description	Why	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

SLA Priority Rule

Type	Child table	Naming	Child row
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What it is. The response and resolution target for one priority, as durations.

How it is used. The heart of the policy. If a ticket's priority has no rule, the SLA is marked **Not Applicable** rather than guessed. Response time may not exceed resolution time.

Field	Label	Type / options	Notes
priority	Priority	Select Low Medium High Critical	required
response_time	Response Time	Duration	required
resolution_time	Resolution Time	Duration	required

Permissions

Child table — permissions are inherited from the parent document.

SLA Working Day

Type	Child table	Naming	Child row
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What it is. One business-hours window: weekday, start time, end time.

How it is used. Multiple windows per day are supported (e.g. a lunch break split). Ignored entirely when the policy is marked 24x7. Deadlines are computed by walking these windows, skipping holidays.

Field	Label	Type / options	Notes
workday	Workday	Select Monday Tuesday Wednesday Thursday Friday Saturday Sunday	required
start_time	Start Time	Time	required; default 09:00:00
end_time	End Time	Time	required; default 18:00:00

Permissions

Child table — permissions are inherited from the parent document.

Service Catalog Document

Type	Child table	Naming	Child row
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What it is. A document the requester must supply for a catalog service, and whether it is mandatory.

How it is used. Used on Service Catalog to tell requesters what to attach.

Field	Label	Type / options	Notes
document_name	Document	Data	required
is_mandatory	Mandatory	Check	default 1
description	Description	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Service Desk Team Member

Type	Child table	Naming	Child row
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What it is. One member of a team: the user, their role in the team (Member / Lead / Backup), their maximum open ticket count and whether they are currently available.

How it is used. `is_available` removes an engineer from the candidate pool entirely (leave, shift).
`max_open_tickets` caps Load Balancing and Skill Based assignment; 0 means unlimited.

Field	Label	Type / options	Notes
<code>user</code>	User	Link → User	required
<code>full_name</code>	Full Name	Data	read-only; fetched from <code>user.full_name</code>
<code>role_in_team</code>	Role In Team	Select Member Lead Backup	default <code>Member</code>
<code>max_open_tickets</code>	Max Open Tickets	Int	default <code>20</code>
<code>is_available</code>	Available	Check	default <code>1</code>

Permissions

Child table — permissions are inherited from the parent document.

Ticket Asset Link

Type	Child table	Naming	Child row
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What it is. An extra ERPNext Asset or configuration item involved in a ticket, beyond the single primary link on the ticket itself.

How it is used. Used when one ticket covers several machines.

Field	Label	Type / options	Notes
<code>asset</code>	Asset	Link → Asset	
<code>asset_name</code>	Asset Name	Data	read-only; fetched from <code>asset.asset_name</code>
<code>configuration_item</code>	Configuration Item	Link → Configuration Item	
<code>notes</code>	Notes	Small Text	

Permissions

Child table — permissions are inherited from the parent document.

Ticket Timeline

Type	Child table	Naming	Child row
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What it is. The human-readable activity feed of a ticket, incident or change: time, event type, user, description and the from/to values of whatever changed.

How it is used. Written by `utils.add_timeline()` from the controllers and every engine. Read-only. This is what the form's History section shows, alongside the audit log.

Field	Label	Type / options	Notes
event_time	Time	Datetime	required
event_type	Event	Select Created Status Change Assignment Comment Approval SLA Attachment Work Log Escalation Linked Record Other	
user	User	Link → User	
description	Description	Small Text	
from_value	From	Data	
to_value	To	Data	

Permissions

Child table — permissions are inherited from the parent document.

Ticket Watcher

Type	Child table	Naming	Child row
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What it is. Someone who is copied on a ticket without owning it, and how much they want to hear - All Updates, Status Change only, or Resolution Only.

How it is used. The notification engine appends watchers to the recipient list of every event they subscribed to, provided `notify_watchers` is on in Service Desk Settings.

Field	Label	Type / options	Notes
user	User	Link → User	required
notify_on	Notify On	Select All Updates Status Change Resolution Only	default All Updates

Permissions

Child table — permissions are inherited from the parent document.

7. The engines — business logic in detail

Seven engine modules under `turacoz_it_service_desk/engine/` carry all the behaviour that is not tied to a single document. The controllers call into them; so do the scheduled jobs and the REST API. If you want to understand why the system did something, this is the chapter.

7.1 SLA engine — `engine/sla.py`

Policy resolution

When a ticket is created (or when its priority or category changes while it is still open), a policy is resolved by taking the **first** match in this order:

1. The **Service Catalog** item's SLA policy, if a service is chosen and the policy is active.
2. The **sub-category**'s Default SLA Policy, if active.
3. An active SLA Policy whose Apply To Category equals the sub-category.
4. The **category**'s Default SLA Policy, if active.
5. An active SLA Policy whose Apply To Category equals the category.
6. **Service Desk Settings → Default SLA Policy**, if active.
7. Any active policy flagged **is_default**.

If nothing resolves, or the resolved policy has no priority rule for this ticket's priority, the SLA status is set to **Not Applicable** rather than guessed. That is the usual explanation for a ticket with blank Respond By and Resolve By dates.

Working-time arithmetic

All SLA maths is done in working time, never wall-clock time — unless the policy is ticked **24x7**, in which case calendar seconds are used directly.

To place a deadline, the engine walks forward day by day from the ticket's opening time, consuming the policy's business-hours windows for each weekday, skipping any date present in the policy's **Holiday List**, until the required number of seconds has been consumed. Several windows per day are supported (for a split shift). The walk is capped at 400 days; if a policy has no reachable business hours at all, the engine falls back to calendar time rather than looping.

The same walk in reverse measures elapsed business seconds, which is what produces the "percentage of SLA consumed" figures on the form and in escalation rules.

Pause and resume

Statuses listed in the policy's **Pause On These Statuses** table (seeded with Pending User and Pending Vendor) stop the clock:

- On entering a pause status: `hold_started_on` is stamped and SLA status becomes **Paused**.
- On leaving it: the working seconds held are added to `total_hold_time`, and **both** deadlines are pushed forward by the same working duration. The response deadline is only pushed if the first response has not yet happened.
- A **Paused / Resumed** pair is written to the SLA Events table with the hold length in minutes.

Milestones

Milestone	Triggered by	Effect
First response	The first engineer comment, the first work log, or the first move into In Progress / Testing / Resolved / User Verification / Closed	Stamps first_responded_on, sets response_sla_met, logs Response Met or Response Breached . A comment written by the requester does not count.
Resolution	Moving into Resolved or Closed	Stamps resolved_on, sets resolution_sla_met, and sets the overall SLA status to Fulfilled only if both halves were met, otherwise Failed .
Reopen	Moving into Reopened from Resolved, Closed or User Verification	A brand-new resolution deadline is computed from now; resolved date and resolution compliance are cleared; SLA status returns to Ongoing . The response half is not re-run.

Breach detection and escalation

Every 15 minutes the scheduler sweeps all tickets that are not Resolved, Closed or Cancelled, whose SLA status is Ongoing / Response Overdue / Resolution Overdue, and which have a policy. For each one:

1. If the response deadline has passed and the first response has not happened, SLA status becomes **Response Overdue**, compliance is set to failed, an SLA Event is written and an `sla_breach` notification fires.
2. If the resolution deadline has passed, the same happens with **Resolution Overdue**.
3. Every escalation rule on the policy is evaluated. A rule fires when the elapsed percentage of its chosen clock (Response or Resolution) reaches its threshold. Firing:
 - sets `is_escalated` and increments `escalation_level`;
 - optionally raises the priority one step (capped at Critical), recording it on the timeline;
 - notifies the target user and/or everyone holding the target role, and optionally the requester;
 - writes an **Escalated** SLA Event tagged `rule:<n>`.

That tag is the idempotency guard — a rule that has already fired on this ticket never fires again.

A failure on one ticket is logged and the sweep continues to the next, so one bad record cannot stall the whole job.

Pre-breach reminders

Hourly, any open ticket whose resolution clock has passed the Send Breach Reminder At (% elapsed) threshold (default 75 %) but has not yet breached gets one `sla_reminder` notification to the engineer and team lead. A marker in the SLA Events table stops it repeating.

The live countdown

`sla.get_sla_status(ticket)` is whitelisted and returns the policy, both deadlines, remaining working seconds on each (negative when overdue), the percentage consumed, total hold time and the escalation level. It powers both the form's SLA indicators and the `api.ticket.sla_status` endpoint.

7.2 Assignment engine — `engine/assignment.py`

Choosing the team

1. A team already on the ticket wins.
2. Otherwise the sub-category's **Default Team**, if active.
3. Otherwise an active team whose **Default Category** is that sub-category.
4. Then the same two checks against the category.

5. Finally **Service Desk Settings** → **Default Team**.

Choosing the engineer

Candidates are the team members who are marked **Available** and whose user account still exists and is enabled. If the category or sub-category names a **Default Engineer** and that person is in the candidate list, they win outright. Otherwise the team's method decides:

Method	How it picks
Round Robin	Finds the engineer on the most recent ticket for this team and takes the next candidate in list order, wrapping around.
Load Balancing	Counts each candidate's currently open tickets, discards anyone at or above their Max Open Tickets (0 = unlimited), and picks the least loaded — ties broken alphabetically for determinism. If everybody is at capacity, it still picks the least loaded rather than leaving the ticket unassigned.
Skill Based	Looks up active Engineer Skill rows for the candidates in this category or sub-category and sorts by: exact sub-category match first, then Expert > Intermediate > Beginner, then certified before uncertified, then lowest open load, then user id. The first candidate under capacity wins. Falls back to Load Balancing if no skill rows exist.
Manual	Assigns nobody. The ticket stays unassigned for a human to pick up.

Applying it

Auto-assignment runs inside the ticket's `validate()`, so it never fires a second save. It is skipped entirely when Auto Assign New Tickets is off in settings, when the ticket is Resolved, Closed or Cancelled, and while approval is pending. A successful assignment moves a Draft or Open ticket to **Assigned** and appends an Assignment History row.

After insert (and on every change of engineer) the assignment is mirrored into Frappe's own **ToDo** / `_assign` mechanism, so the ticket appears in the standard assignment sidebar and in the assignee's Frappe to-do list. Ticket priority is mapped to ToDo priority (Critical and High both become High). A failure here is logged, never raised.

`reassign(ticket, engineer, team, reason)` and `get_workload(team)` are whitelisted for the form's **Reassign** dialog and the manager dashboard.

7.3 Approval engine — `engine/approval.py`

Covered as a workflow in section 5.5. The mechanics worth knowing:

- **Matrix scoring returns None, not zero, on a violated condition** — so a matrix restricted to a department genuinely never applies outside it, rather than merely scoring lower.
- **Level conditions** are evaluated with `frappe.safe_eval` against `doc`. A condition that raises is logged and treated as true (fail-open), so a typo does not silently drop an approval level — but the matrix validation compiles every condition on save, so syntax errors are caught before they reach production.
- `get_approval_state(doctype, name)` is what the form calls to decide whether to render the Approve and Reject buttons; it also returns who the item is currently pending with.
- `get_pending_approvals(user)` scans Approval History across Service Ticket, Change Request and Privileged Access Request, keeping only rows that are pending, whose parent is still pending, and whose level equals the parent's current level — so a parallel matrix does not show you levels you cannot yet act on.
- Approve and reject both write an audit row and fire notifications, then commit.

7.4 Notification engine — `engine/notification.py`

Sixteen templated events, each with a subject and an HTML body, delivered over **e-mail** (`frappe.sendmail` , queued) and **in-app alert** (a Frappe Notification Log entry). A link back to the document is appended to every message.

Event	Fires when	Default recipients
<code>new_ticket</code>	A ticket is created	Assigned engineer, team lead
<code>assignment</code>	A ticket is assigned or reassigned	The new engineer
<code>status_change</code>	Status changes to anything not specially handled	Requester
<code>comment</code>	A Public Note is added	Requester and engineer (only the engineer if the requester wrote it)
<code>approval_pending</code>	An approval trail starts or advances a level	Every resolved pending approver, or everyone holding the pending role
<code>approval_approved</code>	An approve action	Requester and engineer
<code>approval_rejected</code>	A reject action	Requester and engineer
<code>resolution</code>	Status becomes Resolved	Requester
<code>closure</code>	Status becomes Closed	Requester
<code>reopened</code>	Status becomes Reopened	Engineer, team lead
<code>sla_breach</code>	A response or resolution deadline passes	Engineer, team lead
<code>sla_reminder</code>	The pre-breach percentage threshold is crossed	Engineer, team lead
<code>escalation</code>	An escalation rule fires	The rule's user and/or role, team lead, and the requester if the rule says so
<code>feedback_request</code>	Daily job, for tickets closed yesterday with no feedback	Requester
<code>access_granted</code>	Privileged access becomes Approved or Active	Requester and the person it is for
<code>access_expiring</code>	Daily job, one day before expiry	Requester and the person it is for
<code>access_revoked</code>	Access is revoked or expires	Requester and the person it is for

Gates and guarantees

- Nothing is sent if both Email Notifications and System Notifications are off.
- `sla_breach` and `escalation` are additionally gated by Notify On SLA Breach.
- **Watchers** are appended to the recipient list of every event they subscribed to (All Updates / Status Change only / Resolution Only), provided Notify Watchers is on.
- Administrator, Guest, disabled users and users without an e-mail address are dropped. Duplicates are removed.
- **You are not notified of your own action**, except for approval events.
- Every attempt is recorded in the document's **Notifications Sent** table with the channel, recipient, event, timestamp and Sent/Failed plus the error text. **This is the first place to look when someone says they never got an e-mail.**

- **Notification failure never blocks work.** The whole engine is wrapped so that any exception is logged to the Error Log and swallowed — a broken mail server can never stop a ticket from being saved.

7.5 Audit engine — `engine/audit.py`

Registered in `hooks.py` as `after_insert`, `on_update` and `on_trash` document events on twelve doctypes (thirteen are recognised by the engine, which additionally accepts Service Desk Settings when called directly):

Service Ticket, Incident, Problem, Change Request, Privileged Access Request, Knowledge Article, Configuration Item, Work Log, Ticket Comment, SLA Policy, Approval Matrix, Service Desk Team (and Service Desk Settings).

Aspect	Behaviour
What is captured	Timestamp, reference doctype and name, action, acting user and their full name, request IP address, browser user-agent (first 200 chars), and a JSON diff of every changed field as <code>{"field": {"from": ..., "to": ...}}</code> .
Actions recorded	Created, Modified, Assigned, Status Changed, Approved, Rejected, Comment, Attachment, Deleted, Exported, Viewed, Escalated, SLA Event, Access Granted, Access Revoked, API Call.
Action inference	A change touching status is logged as Status Changed ; touching assigned_engineer as Assigned ; touching approval_status as Approved or Rejected ; anything else as Modified .
Noise filter	<code>modified</code> , <code>modified_by</code> , <code>creation</code> , <code>owner</code> , <code>idx</code> , <code>docstatus</code> , <code>tags</code> , <code>comments</code> , <code>_assign</code> , <code>likes</code> , and the timeline / SLA event / notification tables are never diffed. A save that changes nothing else writes no row.
Immutability	The controller throws on any attempt to modify an existing row, and on any delete by a user other than Administrator.
Retention	The daily job deletes rows older than Audit Retention (days). Default 2555 days = 7 years, chosen for ISO 27001 and SOC 2. Setting it to 0 keeps everything forever.
Safety	Writing an audit row never raises — a failure is logged to the Error Log so it can never block a business transaction.
Reading it	<code>audit.get_trail(doctype, name, limit)</code> is whitelisted and permission-checked against the referenced document.

7.6 Automation engine — `engine/automation.py`

The three scheduler entry points and what they do, in order:

Schedule	Job	What it does
<code>* / 15 * * * *</code>	<code>every_fifteen_minutes</code>	The SLA breach and escalation sweep (section 7.1).
hourly	<code>hourly</code>	1. Pre-breach reminders. 2. Privileged access expiry — anything Approved or Active past its valid-till is Revoked (if Auto Revoke is set, with an audit row and a notification) or otherwise marked Expired.

Schedule	Job	What it does
daily	<code>daily</code>	1. Auto close — tickets in Resolved or User Verification whose resolved date is older than Auto Close After (days) are closed, with a timeline note and an audit row. 2. Satisfaction surveys — a feedback request for every ticket closed since yesterday that has no rating and no feedback record. 3. Auto problem creation — see below. 4. Expiring access warnings — one day ahead. 5. Audit purge — retention enforcement.

Auto problem creation

Counts tickets of type **Incident** created in the last 30 days, grouped by category, that are not already linked to a problem. Any category at or above Auto Create Problem After N Similar Incidents (default 5) gets a Problem — either an existing open one for that category, or a new **High** priority problem titled "Recurring incidents in <category>". Every matching ticket is then linked to it and the problem's incident count updated. Disabled by setting the threshold to 0 or turning Problem Management off.

7.7 Knowledge engine — `engine/knowledge.py`

Tokenisation strips HTML, lower-cases, keeps alphanumeric words of 3+ characters and drops a stop-word list of common English filler plus service-desk noise words ("issue", "problem", "error", "help", "unable", ...).

Suggestion scoring

```
score = 5 × (number of words shared between the ticket text and the
            article's title + keywords)
      + 4 if the article's category matches the ticket's category
      + 3 if the article's sub-category matches
      + min(helpful_count, 10)
      - min(not_helpful_count, 5)
```

Articles scoring 0 or less are discarded. Only Published articles are considered. Results are sorted by score descending, then title, and truncated to the limit (default 5).

Other functions: free-text `search()` across title, keywords and content, ordered by helpful count then views then recency; `record_view()` and `rate()` which increment counters without touching the modified timestamp; and `create_article_from_ticket()`, which drafts a **Troubleshooting** article with Problem / Root Cause / Resolution headings from a resolved ticket, derives keywords from the subject, links the ticket both ways and refuses if the ticket has no resolution.

7.8 Shared helpers — `engine/utils.py`

- `setting(fieldname, default)` — the single accessor every engine uses to read Service Desk Settings.
- `derive_priority(impact, urgency)` and `raise_priority(priority)` — the priority matrix and the one-step escalation bump.
- `get_users_with_role(role)` — enabled users holding a role, used for role-based notification and escalation.
- `get_reporting_manager(user)` — user → Employee → `reports_to` → that employee's user id. **This is the single dependency that makes "Requester Manager" approval levels work.**
- `get_department_head(department)` — probes the Department doctype for `department_head`, `head_of_department` or `approver`, whichever exists on this site, and maps it back to a user.
- `add_timeline(...)` — the one way anything writes to a document's timeline.

8. Complete report listing

Twelve **Script Reports** ship with the module. All are registered under the module **Turacoz IT Service Desk**, marked standard, and print with the **THS** letterhead. Open any of them at `/app/query-report/<name>` or from the **Reports** card on the workspace.

8.1 All 12 at a glance

Report	Based on	Answers	Visible to
Service Desk Ticket Summary	Service Ticket	"Show me the tickets." The flat operational list with SLA, age and worked hours.	+ Engineer
SLA Compliance	Service Ticket	"Are we hitting our targets?" Response and resolution compliance, grouped however you like.	Managers
SLA Violations	Service Ticket	"Exactly which promises did we break, by how much?"	Managers
Engineer Performance	Service Ticket	"How is each engineer doing?" Volume, SLA, reopens, effort, rating.	Managers
Ticket Aging	Service Ticket	"What is going stale?" Open tickets bucketed by age.	+ Engineer
Ticket Analysis	Service Ticket	"Where is the volume coming from?" Any dimension you choose.	Managers
Reopened Tickets	Service Ticket	"What did we get wrong the first time?"	Managers
Monthly KPI	Service Ticket	"The management scorecard." One row per month.	Managers
Change Success Report	Change Request	"Are our changes safe?" Success rate by type and risk.	Managers
Incident Report	Incident	"What broke, how bad, how fast did we contain it?"	Managers + CISO
Knowledge Base Usage	Knowledge Article	"Is the knowledge base earning its keep?"	+ Engineer
Top Requesters	Service Ticket	"Who is asking, and for what?"	Managers

"Managers" means System Manager, IT Manager, Service Desk Team Lead, Service Desk Executive and Auditor. "+ Engineer" means those five plus Service Desk Engineer.

8.2 The shared filter set

Nine of the twelve reports use one common filter block, implemented once in `report/report_utils.py`:

Filter	Type	Default
From Date	Date	Today minus one month
To Date	Date	Today (the whole day is included)
Status	Select — all 12 ticket statuses	blank = all
Priority	Select — Low / Medium / High / Critical	blank = all
Ticket Type	Select — Incident / Service Request / Change Request / Problem / Query	blank = all

Filter	Type	Default
Category	Link → IT Service Category	blank = all
Department	Link → Department	blank = all
Assigned Team	Link → Service Desk Team	blank = all
Assigned Engineer	Link → User	blank = all

The most common report complaint is "my report is empty". Nearly always the cause is the date filter: it defaults to the last month only. Widen From Date before assuming anything is broken.

8.3 Report by report

Service Desk Ticket Summary

The operational workhorse: one row per ticket. Columns: **Ticket, Subject, Type, Status, Priority, Category, Requester, Team, Engineer, SLA, Resolve By, Opened, Resolved, Age (h), Worked (h)**. Uses the shared filter set. Start here when someone asks a question about a specific group of tickets.

SLA Compliance

Columns: **<group>, Tickets, Response Met, Response %, Resolution Met, Resolution %, Breached, Avg Resolution (h)**, with a chart. Adds one extra filter, **Group By**, which can be **Priority** (the default), **Category, Team, Engineer, Department** or **Month**. Change the grouping to slice the same numbers a different way rather than running a different report.

SLA Violations

Columns: **Ticket, Subject, Priority, Status, Engineer, Team, Breach Type, Due, Actual, Overdue By (h), Escalated**. One row per breach, not per ticket — a ticket that missed both its response and its resolution target appears twice, once for each. Still-open breaches are measured against "now", so the overdue hours keep growing until the ticket is dealt with.

Engineer Performance

Columns: **Engineer, Open, Resolved, Reopened, Reopen %, SLA Met, SLA %, Avg Resolution (h), Logged Hours, Avg Rating**. Logged hours come from Work Log; the rating comes from Ticket Feedback via the ticket. Read Reopen % and Avg Rating alongside SLA % — speed on its own is not quality.

Ticket Aging

Columns: **Ticket, Subject, Status, Priority, Engineer, Opened, Age (days), Bucket, SLA**, with a chart of the bucket distribution. Buckets: **0-1 day, 1-3 days, 3-7 days, 7-14 days, 14-30 days, 30+ days**. The shape of the histogram is the health of the backlog.

Ticket Analysis

Columns: **<dimension>, Tickets, Share %, Open, Resolved, SLA %, Avg Resolution (h), Avg Rating**, with a chart. Adds a **Dimension** filter: **Category** (default), **Sub Category, Department, Ticket Type, Source, Priority, Service** or **Configuration Item**. This is the demand-analysis report — use it to find what to automate, document or fix permanently.

Reopened Tickets

Columns: **Ticket, Subject, Category, Engineer, Reopens, Last Reopened, Status, Resolution Category, Rating**. The quality report. A cluster in one category usually points at a missing knowledge article or an unresolved underlying problem.

Monthly KPI

Columns: **Month, Raised, Resolved, Closed, Backlog, SLA %, First Response %, Reopen %, Avg Resolution (h), Avg Rating, Escalations**, with a chart. One row per calendar month. Note that the star ratings are rescaled to a 0–5 figure for reporting.

Change Success Report

Columns: **Change Type, Risk, Total, Successful, With Issues, Failed, Rolled Back, Success %, Avg Duration (h)**, with a chart. Filters: **From Date, To Date, Change Type**. Counts only changes that reached Completed, Closed or Rolled Back; "success" means a validation result of Successful or Successful with Issues.

Incident Report

Columns: **Incident, Title, Type, Severity, Status, Security, Affected Service, Users Hit, Detected, Contained (h), Resolved (h), Manager**. Filters: **From Date, To Date, Severity, Incident Type**, and a **Security Only** checkbox. Contained and Resolved hours are measured from the detection time. This is the report the CISO lives in.

Knowledge Base Usage

Columns: **Article, Title, Type, Category, Status, Views, Helpful, Not Helpful, Helpful %, Used On Tickets, Version**. Single filter: **Category**. "Used On Tickets" counts tickets whose Knowledge Article Used field points at the article — that is the column that proves deflection. A high-view, low-helpful article is a rewrite candidate.

Top Requesters

Columns: **Requester, Department, Tickets, Share %, Incidents, Requests, Open, Avg Rating**. Uses the shared filter set. Useful for spotting a department that needs training, or a single user whose repeated requests should become a self-service item.

9. Desk experience

9.1 The workspace

One public workspace named **IT Service Desk** (icon tool, blue, sequence 40), built by `setup/desk.py` and version-controlled as the app's only fixture. It is laid out as: a header, then the seven shortcuts, then a **Key Numbers** header with all eight number cards, then an **Analytics** header with the first four charts, then a **Masters & Reports** header with the six link cards.

Shortcuts (7)

Shortcut	Goes to	Live count shows
Service Ticket	Service Ticket list	All tickets not Resolved / Closed / Cancelled
My Open Tickets	Service Ticket list	Open tickets assigned to you
Incident	Incident list	—
Change Request	Change Request list	—
Knowledge Article	Knowledge Article list	—
SLA Compliance	The report	—
Service Desk Settings	The settings single	—

Number cards (8)

Card	Counts
Open Tickets	Tickets not Resolved / Closed / Cancelled
Critical Open Tickets	... and priority Critical
SLA Breached	Tickets with SLA status Failed, Response Overdue or Resolution Overdue
Unassigned Tickets	Open tickets with no engineer
Escalated Tickets	Open tickets flagged escalated
Open Incidents	Incidents not Resolved / Closed
Changes Awaiting Approval	Change Requests with approval status Pending
Active Privileged Access	Access requests Approved or Active

Charts (5)

Chart	Type	Shows
Tickets by Priority	Donut	Open tickets grouped by priority
Tickets by Status	Bar	Open tickets grouped by status
Tickets by Category	Bar	All tickets, top 10 categories
Ticket Volume Trend	Line	Weekly ticket count over the last quarter
Incidents by Severity	Bar	All incidents grouped by severity

Link cards (6, covering 36 links)

Card	Contains
Tickets	Service Ticket, Work Log, Ticket Comment, Ticket Feedback
Incidents, Problems & Changes	Incident, Problem, Change Request
Access & Audit	Privileged Access Request, Service Desk Audit Log
Knowledge & CMDB	Knowledge Article, Configuration Item
Configuration	Service Desk Settings, Service Desk Team, SLA Policy, Approval Matrix, IT Service Category, Service Catalog, Engineer Skill
Reports	All 12 script reports

9.2 The ticket list view

- Opens pre-filtered to hide Closed and Cancelled tickets.
- Each row is coloured by status: grey Draft, orange Open, blue Assigned and In Progress, yellow Pending, purple Testing, green Resolved and Closed, light blue User Verification, red Reopened.
- **Any open ticket whose SLA has failed or gone overdue is shown as a red "SLA Breached" pill instead of its status colour** — so breaches cannot be missed while scrolling.
- Priority is rendered as a coloured pill: red Critical, orange High, blue Medium, grey Low.

9.3 Ticket form buttons

Button	Group	Does
Approve / Reject	top level	Shown only when the server confirms you are a pending approver at the current level. Reject demands a reason.
Reassign	Actions	Dialog for engineer, team and reason. Records assignment history and a timeline entry.
Add Work Log	Actions	Logs effort against the ticket.
Suggest Articles	Actions	Runs the knowledge suggestion engine against this ticket's text and shows the ranked matches.
Incident / Problem / Change Request	Create	Creates the ITIL record pre-linked to this ticket in both directions.
Create Knowledge Article	Create	Drafts a Troubleshooting article from the resolution. Requires a resolution.

The form also renders the live SLA indicators (both countdowns, percentages consumed, hold time, escalation level) fetched from `sla.get_sla_status`.

10. Dashboards

Four role-shaped dashboard endpoints in `api/dashboard.py` , plus a combined `metrics` call. All are whitelisted and respect permissions.

Endpoint	Arguments	Returns
<code>dashboard.employee</code>	<code>user</code> (defaults to you)	open, pending_with_me (Pending User + User Verification), resolved_this_month, awaiting, most recent tickets, and the 5 most-viewed published articles.
<code>dashboard.engineer</code>	<code>user</code>	assigned, in_progress, due_today, overdue, resolved_this_month, sla_compliance %, average hours_this_week from Work Log, and a 15-row queue sorted Critical→Low then by nearest c
<code>dashboard.manager</code>	<code>team</code> , <code>days</code> (default 30)	open_total, critical_open, escalated, breached, unassigned, resolved, sla_compliance, response reopen_rate, plus breakdowns by status, priority, top 10 categories, top 10 departments, a workload.
<code>dashboard.executive</code>	<code>months</code> (default 6)	Monthly trend of raised / resolved / SLA met; incidents by severity; security incident count; changes completed; customer satisfaction and feedback count; open problems; active private article count.
<code>dashboard.metrics</code>	—	The employee, engineer and manager payloads in one call, for the workspace.

11. REST API reference

11.1 Calling convention

```
Base    https://<site>/api/method/turacoz_service_desk.turacoz_it_service_desk.api.<module>.<method>
Auth    Authorization: token <api_key>:<api_secret>
Verb    POST for anything that writes, GET for reads
```

Every endpoint returns the same envelope:

```
success  {"success": true, "data": <payload>, "message": "Ticket created"}
failure  {"success": false, "error": {"code": "validation_error", "message": "..."} }
```

Error code	HTTP	Meaning
permission_denied	403	The token's user may not do this
not_found	404	No such document
validation_error	417	A business rule refused the request — the message is the human-readable reason
server_error	500	Unexpected. The traceback is in the Error Log under "Service desk API failed"

Nothing in the API bypasses permissions. Every endpoint runs as the token's user and performs an explicit permission check. The transaction is rolled back on any error before the envelope is returned.

11.2 `api.ticket` — 15 endpoints

Method	Arguments	Notes
create	Any writable field; <code>subject</code> required; <code>category</code> or <code>service</code> required. Optional <code>watchers</code> list.	Unknown fields are ignored, so clients may post loosely. Source defaults to API , status to Open .
update	<code>ticket</code> + writable fields	Only fields in the writable allow-list are applied. No-op returns "Nothing to update".

Method	Arguments	Notes
<code>get</code>	<code>ticket</code> , <code>with_activity</code> (default 1)	Full payload; with activity it also returns comments, work logs, timeline and approvals.
<code>list_tickets</code>	<code>status</code> (comma-separated allowed), <code>priority</code> , <code>assigned_engineer</code> , <code>requester</code> , <code>category</code> , <code>team</code> , <code>limit</code> (20), <code>start</code> , <code>order_by</code>	Returns rows plus a <code>total</code> . Honours the ticket visibility rule.
<code>set_status</code>	<code>ticket</code> , <code>status</code> , optional <code>resolution</code> , <code>resolution_category</code> , <code>comment</code>	Subject to the transition matrix.
<code>close</code>	<code>ticket</code> , optional <code>resolution</code> , <code>feedback_rating</code> , <code>feedback_comments</code>	Routes through Resolved first if needed, so the SLA is evaluated correctly.
<code>reopen</code>	<code>ticket</code> , optional <code>reason</code>	Only from Resolved, Closed or User Verification. The reason becomes a public comment.
<code>add_comment</code>	<code>ticket</code> , <code>content</code> , <code>comment_type</code>	Public Note or Internal Note.
<code>comments</code>	<code>ticket</code> , <code>include_internal</code>	Internal notes are hidden automatically from callers without a service desk role.
<code>add_work_log</code>	<code>ticket</code> , <code>description</code> , <code>start_time</code> , optional <code>end_time</code> , <code>activity_type</code> , <code>is_billable</code> , <code>is_internal</code>	Returns the computed hours.
<code>work_logs</code>	<code>ticket</code>	All logs, oldest first.

Method	Arguments	Notes
submit_feedback	ticket, rating, optional comments, response_speed, resolution_quality, engineer_courtesy	One per ticket; a second attempt is refused.
sla_status	ticket	The live countdown payload.
suggest_articles	subject / description / category, or just ticket; limit	Ranked suggestions.
attachments	ticket	Files attached to the ticket.

11.3 api.itsm — 12 endpoints

Method	Arguments	Notes
create_incident	title, incident_type, severity, plus optional service, users, impact, manager, ticket	Refused if Incident Management is disabled. Links back to the ticket.
get_incident	incident	—
create_problem	title, category, priority, description, ticket	Refused if Problem Management is disabled. Records the ticket as Caused By .
create_change	title, change_type, risk; requires business_justification, implementation_plan, rollback_plan	Refused if Change Management is disabled. Created as Draft.
submit_change	change	Moves Draft → Submitted, which starts approval.
request_access	request_type, valid_from, valid_till, business_justification, optional access_scope, requested_for, ticket	If no scope is given, one Admin-level row for the request type is created. Immediately moved to Pending Approval.
revoke_access	request, reason	Reason is mandatory.
approve	doctype, name, comments	Works on Service Ticket, Change Request, Privileged Access Request.

Method	Arguments	Notes
<code>reject</code>	<code>doctype</code> , <code>name</code> , <code>comments</code>	Comments mandatory.
<code>pending_approvals</code>	—	Everything waiting on the calling user, newest first.
<code>search_knowledge</code>	<code>query</code> , <code>category</code> , <code>article_type</code> , <code>limit</code>	Published articles only.
<code>get_article</code>	<code>article</code>	Increments the view counter as a side effect.

11.4 `api.dashboard`

`employee`, `engineer`, `manager`, `executive`, `metrics` — see section 10.

11.5 Examples

```
# raise a ticket
curl -X POST "https://erp.turacoz.com/api/method/turacoz_service_desk\
.turacoz_it_service_desk.api.ticket.create" \
  -H "Authorization: token $KEY:$SECRET" \
  -d subject="VPN is down" -d category="VPN" -d ticket_type="Incident" -d urgency="High"

# live SLA state
curl "https://erp.turacoz.com/api/method/turacoz_service_desk\
.turacoz_it_service_desk.api.ticket.sla_status?ticket=TKT-2026-000001" \
  -H "Authorization: token $KEY:$SECRET"

# approve a change
curl -X POST "https://erp.turacoz.com/api/method/turacoz_service_desk\
.turacoz_it_service_desk.api.itsm.approve" \
  -H "Authorization: token $KEY:$SECRET" \
  -d doctype="Change Request" -d name="CHG-2026-00007" -d comments="Approved for Sunday window"

# manager dashboard
curl "https://erp.turacoz.com/api/method/turacoz_service_desk\
.turacoz_it_service_desk.api.dashboard.manager" \
  -H "Authorization: token $KEY:$SECRET"
```

12. The employee portal at /support

A single self-service page served from www/support/, registered in the standard portal menu as **IT Service Desk** for the **Employee** role. Never cached. Guests are refused with a login prompt.

The page is loaded with:

- Every **active IT Service Category**, ordered by parent then name, so the picker renders as a tree.
- Every **active Service Catalog** item with its category, sub-category, default priority, approval flag and description.
- The signed-in user's **20 most recently updated tickets**.
- Two counters: **open** tickets, and tickets **awaiting the user** (Pending User or User Verification).
- The **six most-viewed published knowledge articles**.

From here an employee can raise a request with live article suggestions, track their tickets and search the knowledge base.

Known gap. Attachment upload and in-portal ticket conversation are not yet exposed on the portal page, although both are fully supported through the REST API.

13. Scheduled jobs

Three entries registered in `hooks.py`. **If the Frappe scheduler is disabled, none of them run** — SLA never escalates, tickets never auto-close, and privileged access is never revoked. Confirm with `bench --site <site> scheduler status`.

Cadence	Method	Work performed
<code>*/15 * * * *</code>	<code>engine.automation.every_fifteen_minutes</code>	SLA breach detection and the escalation ladder.
hourly	<code>engine.automation.hourly</code>	Pre-breach reminders; privileged access revocation and expiry.
daily	<code>engine.automation.daily</code>	Auto-close resolved tickets; satisfaction surveys; auto problem creation; expiring-access warnings; audit log purge.

14. Attached entities — everything outside the module

The module declares `required_apps = ["frappe/erpnext"]` and links into ERPNext and Frappe core rather than duplicating them. It does **not** modify ERPNext core, the `tms` app, or any existing DocType, and it owns no custom fields, property setters or client scripts.

14.1 ERPNext DocTypes linked to

Entity	Linked from	What it is used for
Employee	Service Ticket → <code>requester_employee</code>	Auto-matched from the requester's user id (Active employees only). Supplies the ticket's department when it is blank. Critically, <code>Employee.reports_to</code> is what resolves every "Requester Manager" approval level — missing reporting lines break approval.
Department	Service Ticket, Approval Matrix, Service Catalog	Routing and reporting dimension; matrix matching condition; source of the Department Head approver. The engine probes for <code>department_head</code> , <code>head_of_department</code> or <code>approver</code> , whichever the site has.
Asset	Service Ticket → <code>linked_asset</code> ; Ticket Asset Link; Configuration Item → <code>linked_asset</code>	Ties a ticket to the physical asset it concerns, and lets a CMDB record mirror an ERPNext Asset so finance and operations agree on the same hardware.
Project	Service Ticket → <code>linked_project</code>	Associates service desk effort with a project for costing and reporting.
Cost Center	Service Catalog → <code>cost_center</code>	Charges a catalog service to the right cost centre; pairs with Estimated Cost.
Supplier	Configuration Item → <code>vendor</code>	Records who supports or supplied a configuration item, alongside the warranty / AMC expiry date.
Holiday List	SLA Policy → <code>holiday_list</code>	Directly changes every SLA deadline. Dates present in the list are skipped entirely when working time is counted. Leaving it blank means holidays are treated as ordinary working days.

14.2 Frappe framework DocTypes used

Entity	How the module uses it
User	The identity for every requester, engineer, approver, watcher, owner and auditor. Disabled users are filtered out of assignment candidates and notification recipients.
Role / Has Role	Approval levels, escalation targets and report visibility are all expressed as roles. <code>get_users_with_role()</code> expands a role to its enabled holders for notification.
ToDo (and <code>_assign</code>)	Every ticket assignment is mirrored into Frappe's native assignment mechanism so the ticket appears in the standard sidebar and to-do list. Ticket priority maps to ToDo priority.
Notification Log	The in-app alert channel. Every notification the engine sends creates one of these per recipient, unless system notifications are switched off.
Email Queue / Email Account	E-mail is queued through <code>frappe.sendmail</code> . Without an enabled outgoing Email Account, e-mail delivery is recorded as Failed in Notifications Sent — in-app alerts still work.

Entity	How the module uses it
File	Attachments on tickets, work logs, comments, knowledge articles and RCA documents; exposed through <code>api.ticket.attachments</code> .
Error Log	Every swallowed failure — notification, audit write, API exception, SLA sweep error — is logged here with a title beginning "Service desk".
DocType	Target of the Dynamic Links used by Related Record and Service Desk Audit Log, which is what lets them point at any document in the system.
Module Def	Created and re-created on every migrate so the module survives a fresh clone.
Workspace	The one desk artefact exported as a fixture, filtered to this module only.
Report, Number Card, Dashboard Chart	Created idempotently by <code>setup/desk.py</code> rather than shipped as fixtures.
Holiday	Read directly (as child rows of the Holiday List) when the SLA engine tests whether a date is a working day.

Why fixtures are so tightly filtered. The app exports exactly one fixture — the Workspace, filtered to `module = "Turacoz IT Service Desk"`. An unfiltered fixture entry (a bare `"Custom Field"`, for example) would export every such record from the site it was exported on — including other teams' customisations — and `bench migrate` would then write that snapshot over the target site, silently reverting their work. Keep every future fixture entry filtered.

14.3 Naming-collision note

The category master is called **IT Service Category**, not "Service Category". The shorter name is already taken by the existing Turacoz Management System module on the same site. The preflight check verifies there are no other clashes before installation.

15. Installation and bootstrap

The full production runbook lives in `INSTALL.md` in the app root. This section is the operational summary — enough to install, verify and roll back.

15.1 What the package installs

Object	Count	Notes
DocTypes	36	18 transactional / master + 18 child tables
Roles	7	Existing roles are reused, never overwritten
Script reports	12	—
Number cards / charts	8 / 5	On the IT Service Desk workspace
Workspace	1	IT Service Desk
Seed data	—	60 service categories, 16 catalog services, 1 SLA policy, 1 team, 3 approval matrices
Portal page	1	<code>/support</code>
Scheduled jobs	3	15-minute, hourly, daily

15.2 Prerequisites

- Frappe **15.x** and ERPNext **15.x** on the target site; Python $\geq$ **3.10**; MariaDB **10.6+**; at least 2 GB free disk.
- The **scheduler must be enabled** — SLA escalation, auto-close and access expiry are all scheduled jobs.
- An enabled outgoing **Email Account** if e-mail notification is wanted (in-app alerts work without one).
- A **full backup with files**, copied off the server, before anything else.

15.3 Install

```
cd /home/frappe/frappe-bench

# 1. add the app to the bench (no database change yet)
bench get-app /path/to/turacoz_service_desk

# 2. read-only preflight – checks for DocType / Report / Workspace name clashes,
#   pre-existing roles, missing ERPNext dependencies, scheduler, e-mail, holiday list
bench --site erp.turacoz.com execute \
    turacoz_service_desk.turacoz_it_service_desk.setup.preflight.check
#   expected tail: "No blockers. The app can be installed on this site."

# 3. install
bench --site erp.turacoz.com install-app turacoz_service_desk
bench --site erp.turacoz.com migrate
bench build --app turacoz_service_desk
bench --site erp.turacoz.com clear-cache
bench restart
```

If the preflight lists any blocker, stop. Do not continue until it is resolved.

15.4 What `install-app` does, in order

1. `before_install` — creates the 7 roles. They must exist before the DocType JSON is imported, because the permission rows reference them.
2. Frappe creates the Module Def and imports all 36 DocType JSON files and the 12 report definitions.
3. `after_install` — seeds categories, the **Standard IT SLA** policy, the **IT Service Desk** team, the 3 approval matrices, 16 catalog services and Service Desk Settings, then builds the workspace, number cards and charts.

Everything in the setup layer is idempotent, so any step can be re-run safely. `after_migrate` re-creates the Module Def and the roles on every migrate, so a fresh clone never loses them.

Individual bootstrap commands

```
# roles, categories, SLA policy, teams, approval matrices, catalog, settings
bench --site <site> execute turacoz_service_desk.turacoz_it_service_desk.setup.install.after_install

# reports, number cards, dashboard charts and the workspace
bench --site <site> execute turacoz_service_desk.turacoz_it_service_desk.setup.desk.run

# optional demo data for UAT, and its removal
bench --site <site> execute turacoz_service_desk.turacoz_it_service_desk.setup.demo.seed
bench --site <site> execute turacoz_service_desk.turacoz_it_service_desk.setup.demo.purge
```

15.5 What gets seeded

Seed	Contents
60 categories	14 parents — Infrastructure, ERP, Website, Development, Cloud, Database, Security, Networking, Office365, HRMS, Finance, CRM, Software, Others — with their sub-categories.
Standard IT SLA	Default and active. Targets: Critical 30 min / 4 h, High 1 h / 8 h, Medium 4 h / 24 h, Low 8 h / 72 h. Business hours Mon–Fri 09:00–18:00. Pauses on Pending User and Pending Vendor. Three escalation rules: Response 75 % → Team Lead; Resolution 75 % → Team Lead; Resolution 100 % → IT Manager, raising priority and notifying the requester.
IT Service Desk team	Load Balancing, with Administrator as the initial lead and sole member. Replace this immediately.
3 approval matrices	As listed in section 5.5.
16 catalog services	New Employee Account, Password Reset, Office365 License, VPN Access, New Laptop, Software Installation, Firewall Rule Change, DNS Update, SSL Installation, Website Update, ERP Report Request, Custom Development, API Integration, Database Restore, Cloud Server Provisioning, Backup Restore. Eleven of them require approval.

15.6 Verification

```
bench --site erp.turacoz.com console

frappe.get_installed_apps() # includes turacoz_service_desk
frappe.db.count("DocType", {"module": "Turacoz IT Service Desk"}) # 36
frappe.db.count("Report", {"module": "Turacoz IT Service Desk"}) # 12
frappe.db.count("IT Service Category") # 60
frappe.db.count("Service Catalog") # 16
frappe.db.exists("Workspace", "IT Service Desk") # "IT Service Desk"
frappe.db.get_value("SLA Policy", {"is_default": 1}, "name") # "Standard IT SLA"
```

Then in the browser: the workspace at `/app/it-service-desk`; a new ticket at `/app/service-ticket/new` (check that priority derives from impact × urgency, a team is assigned and both SLA dates appear); the portal at `/support` as a normal employee; the **SLA Compliance** report; and the audit log, where the new ticket's **Created** row must be present and must refuse to be edited.

15.7 Upgrade and rollback

```
# upgrade
bench --site erp.turacoz.com backup --with-files
cd apps/turacoz_service_desk && git pull
cd /home/frappe/frappe-bench
bench --site erp.turacoz.com migrate && bench build --app turacoz_service_desk && bench restart

# rollback, before users have entered data
bench --site erp.turacoz.com uninstall-app turacoz_service_desk --yes
bench --site erp.turacoz.com clear-cache && bench restart

# rollback, after users have entered data – restore the backup
bench --site erp.turacoz.com --force restore \
  sites/erp.turacoz.com/private/backups/<timestamp>-erp_turacoz_com-database.sql.gz
```

`uninstall-app` drops the module's DocTypes **and all their data**. The 7 roles and any Holiday List you touched remain and must be removed by hand if required.

To switch the module off without removing it: disable the scheduled jobs and strip the roles from users. The DocTypes stay, but nothing runs.

Schema changes ship as DocType JSON and are applied by `migrate`. Data changes ship as patches listed in `patches.txt` — currently one, `v1_0.fix_dashboard_filter_format`, which repairs number card and chart filters written in the short form by version 1.0.0.

16. Configuration checklist for a new site

Do all eight of these **before** announcing the system to users. Items 1–4 change behaviour immediately; skipping item 5 is the single most common cause of a broken go-live.

#	Where	What to do
1	<code>/app/service-desk-settings</code>	Confirm the default SLA policy and default team. Switch off any module you are not using (Change, Problem, CMDB, Privileged Access) — this also stops the related jobs and API endpoints. Set Auto Close After (days), the breach reminder percentage and the audit retention period.
2	<code>/app/service-desk-team/IT Service Desk</code>	Replace Administrator as the team lead. Add your engineers as members, set each one's Max Open Tickets, and pick the assignment method.
3	User records	Assign the service desk roles to the right people. Every employee already holds Employee , which is enough to raise tickets. Do not hand out Service Desk Engineer casually — it grants visibility of every ticket in the system.
4	<code>/app/sla-policy/Standard IT SLA</code>	Review the four priority targets. Set the Holiday List so SLA working time excludes public holidays. Adjust the business-hours rows, or tick 24×7 for a follow-the-sun desk.
5	Approval matrices & Employee records	The seeded matrices use Requester Manager , which resolves through <code>Employee.reports_to</code> . Confirm the reporting lines exist , otherwise mandatory approval levels fail with "No approver could be resolved" and requesters cannot submit. Also confirm somebody actually holds IT Manager and CISO .
6	<code>/app/engineer-skill</code>	Only needed if any team uses Skill Based assignment. Without skill rows it silently falls back to Load Balancing.
7	Categories and catalog	Prune or extend the 60 seeded categories and 16 services to match how your organisation actually works. Set each category's default team, default engineer, default SLA policy and approval flag — this is where routing is really configured.
8	<code>/me</code> as a normal employee	Verify the IT Service Desk portal entry appears and <code>/support</code> loads.

17. Testing and demo data

17.1 The test suite

```
bench --site <site> run-tests --skip-before-tests \  
--module turacoz_service_desk.turacoz_it_service_desk.doctype.service_ticket.test_service_ticket
```

Expect **18 tests, all passing**. They cover the priority matrix, working-hour deadline arithmetic including the weekend skip, SLA pause and resume, first response, resolution compliance, reopen, the status-transition guards, category routing, load balancing, the audit trail and approval matrix scoring.

Run the suite on a UAT copy if you possibly can. If you must run it on production, the `--skip-before-tests` flag is **mandatory** — without it Frappe executes ERPNext's `before_tests` hook, which creates a test company, fiscal year and other setup records on the live site. The tests themselves insert inside a transaction and roll back.

Test files ship next to each doctype (`test_<doctype>.py`) for Service Ticket, SLA Policy, Approval Matrix, Change Request, Incident, Problem, Privileged Access Request, Configuration Item, Knowledge Article, Service Catalog, Service Desk Team, Service Desk Settings, Service Desk Audit Log, Engineer Skill, IT Service Category, Ticket Comment, Ticket Feedback and Work Log.

17.2 Demo data

```
bench --site <uat-site> execute turacoz_service_desk.turacoz_it_service_desk.setup.demo.seed  
bench --site <uat-site> execute turacoz_service_desk.turacoz_it_service_desk.setup.demo.purge
```

`seed` creates 60 tickets spread over 90 days, walked through realistic status histories, plus incidents, changes, a problem, knowledge articles and feedback, so the dashboards and reports have something to show during UAT. `purge` removes exactly what it created — tickets tagged `demo` and records prefixed `[DEMO]` .

Never run the demo seeder on production.

18. Troubleshooting

Symptom	Cause and fix
SLA deadlines are blank; SLA status says Not Applicable	No policy resolved, or the resolved policy has no priority rule for this ticket's priority. Check Service Desk Settings → Default SLA Policy, and that the category or service points at an active policy. Also check the policy really has a rule for that priority.
SLA never escalates; tickets never auto-close; access is never revoked	The scheduler is disabled, or <code>pause_scheduler</code> is set in the site config. Run <code>bench --site <site> scheduler status</code> .
Deadlines look far too generous or too tight	Working-time arithmetic. A Medium ticket with a 24-hour resolution target on a Mon-Fri 09:00–18:00 policy is due three working days later, not tomorrow. Either accept that, or mark the policy 24×7.
Deadlines ignore public holidays	No Holiday List is set on the SLA Policy.
Tickets are not being assigned	Check, in order: Auto Assign New Tickets is on; the team is active; members are marked Available ; the assignment method is not Manual ; members are not all at their Max Open Tickets; approval is not still pending (routing is skipped while it is).
Skill Based assignment behaves like Load Balancing	No active Engineer Skill rows match the ticket's category or sub-category. That fallback is deliberate.
"No approver could be resolved for approval level N"	A mandatory level cannot find anyone. For Requester Manager , the requester needs an Active Employee record with <code>reports_to</code> set, and that manager needs a <code>user_id</code> . For a role level, somebody must actually hold the role.
Approve / Reject buttons do not appear	You are not a pending approver at the current level. In a Sequential matrix only the lowest pending level can act.
A ticket refuses to move to In Progress	Either approval is still Pending, or the move is not in the transition matrix (section 5.3).
"Add a resolution before marking the ticket Resolved"	Working as designed — the Resolution field is mandatory for Resolved and Closed.
E-mails are not delivered	No enabled outgoing Email Account. Open the ticket's Notifications Sent table — failed sends are recorded there with the error. In-app alerts still work.
Users are notified about their own actions	They should not be, except for approval events. If it happens elsewhere, check whether the user is also listed as a watcher.
Employees can see other people's tickets	They hold a service desk role. Visibility is granted by role first, then by requester / requested-for / owner / assignee.
A report is empty	Widen the From/To dates — they default to the last month only.
Workspace, cards or charts missing	<code>bench --site <site> execute turacoz_service_desk.turacoz_it_service_desk.setup.desk.run</code>
Red popup "Invalid filter: not in" on the workspace	Number card / chart filters written in the short form by version 1.0.0. Fixed in 1.0.1 — run <code>bench migrate</code> , or <code>setup.desk.run</code> , then <code>clear-cache</code> and reload the browser.
"My Open Tickets" always shows 0	The same 1.0.0 defect (a literal <code>"user"</code> in the shortcut filter). Same fix.
Seed data missing (no categories or SLA policy)	Re-run <code>setup.install.after_install</code> . It is idempotent.
Module import failed ... No module named <code>'tms.turacoz_it_service_desk'</code>	Stale cache from an earlier layout, when the module lived inside the <code>tms</code> app. <code>bench --site <site> clear-cache && bench restart</code>

Symptom	Cause and fix
Audit log rows cannot be edited or deleted	Working as designed. Only Administrator may delete, and nobody may edit.
A Configuration Item will not retire	Open tickets still reference it. The error names the count.
Something silently did not happen	Look in <code>/app/error-log</code> for entries titled "Service desk...". Notification, audit and API failures are logged there rather than raised, by design.

19. Source code map

turacoz_service_desk/	
├─ INSTALL.md	production runbook
├─ README.md	overview
├─ pyproject.toml	flit packaging (version 1.0.1)
├─ turacoz_service_desk/	
│ ├─ hooks.py	doc events, scheduler, permissions, portal menu, fixtures
│ ├─ modules.txt	"Turacoz IT Service Desk"
│ ├─ patches.txt	data patches applied by bench migrate
│ ├─ fixtures/workspace.json	the one exported fixture
│ ├─ patches/v1_0/	fix_dashboard_filter_format.py
│ ├─ www/support/	employee self-service portal (index.py + index.html)
│ └─ turacoz_it_service_desk/	
│ └─ _build_doctypes.py	one-shot schema builder (the JSON remains the source of truth)
│ └─ api/	
│ └─ response.py	the {"success": ...} envelope and the @api decorator
│ └─ ticket.py	15 ticket endpoints
│ └─ itsm.py	12 incident/problem/change/access/knowledge endpoints
│ └─ dashboard.py	employee · engineer · manager · executive · metrics
│ └─ doctype/	36 DocTypes: .json schema, .py controller, .js client script, test_*.py
│ └─ engine/	
│ └─ sla.py	policy resolution, working-time maths, pause/resume, breach detection, escalation, live countdown
│ └─ assignment.py	team routing, round robin / load / skill, ToDo sync
│ └─ approval.py	matrix scoring, level resolution, approve & reject
│ └─ notification.py	16 templated events over e-mail and in-app alert
│ └─ audit.py	append-only audit trail and retention
│ └─ automation.py	the three scheduler entry points
│ └─ knowledge.py	tokenising, suggestion scoring, search, article creation
│ └─ utils.py	settings accessor, priority matrix, role/manager lookups, timeline writer
│ └─ report/	
│ └─ report_utils.py	the shared filter block and column helper
│ └─ <12 report folders>	.py query, .js filters, .json definition
│ └─ setup/	
│ └─ preflight.py	read-only pre-install safety check
│ └─ install.py	roles, module def, categories, SLA, team, matrices, catalog
│ └─ desk.py	reports, number cards, charts, workspace
│ └─ demo.py	UAT seed data and its purge

Where to look when you need to change something

To change...	Edit
Allowed ticket status transitions	doctype/service_ticket/service_ticket.py → ALLOWED_TRANSITIONS
The impact x urgency priority matrix	engine/utils.py → PRIORITY_MATRIX
Notification wording	engine/notification.py → TEMPLATES
Which doctypes are audited	hooks.py → AUDITED_DOCTYPES and engine/audit.py
Job cadence	hooks.py → scheduler_events
Knowledge suggestion scoring	engine/knowledge.py → suggest()
Report columns or filters	The report's .py (columns and query) and .js (filters)
Workspace layout, cards, charts	setup/desk.py , then re-run desk.run
Seeded master data	setup/install.py

19.1 Known gaps and planned increments

- **Teams, SMS and WhatsApp notification channels.** The Notification Sent table already carries the channel field; only **Email** and **System** are wired up.
- **Integrations:** Azure AD / Microsoft Graph, Jira import, GitHub, Slack.
- **CAB calendar view** and a change-freeze calendar.
- **Portal attachment upload** and in-portal ticket conversation (both are already available through the REST API).

20. Glossary

Term	Meaning in this system
Approval Matrix	A reusable, condition-driven approval ladder shared across tickets, changes, access requests and catalog items.
Auto close	The daily job closing tickets left in Resolved or User Verification for longer than the configured number of days.
Breach	A response or resolution deadline that has passed without being met.
CAB	Change Advisory Board — the group that approves risky changes. Forced on for High/Critical risk and Emergency changes.
CI / CMDB	Configuration Item — a tracked piece of infrastructure; the Configuration Management Database is the collection of them and their relationships.
CAPA	Corrective and Preventive Action, recorded on an Incident after root cause analysis.
Escalation level	How many escalation rules have fired on a ticket. Increments each time; never resets.
First response	The first engineer activity on a ticket — comment, work log, or move into a working status. A requester's own comment does not count.
Hold time	Working time a ticket spent in a pausing status. Added back onto both deadlines when the ticket resumes.
Impact	How much of the business is affected. Combined with urgency to derive priority.
Internal Note	A comment visible only to service desk roles — never to the requester or on the portal.
Known Error	A problem whose root cause is understood and for which a workaround exists, but which is not yet permanently fixed.
Priority	Low / Medium / High / Critical. Normally derived from impact × urgency, but a user's explicit choice is respected.
Public Note	A comment the requester can see and is e-mailed about.
RCA	Root Cause Analysis. Mandatory before closing an S1 or S2 incident, and before resolving a problem.
Reopen	A requester rejecting a resolution. Restarts the resolution clock and counts against the reopen rate.
Requester Manager	An approver type resolved through <code>Employee.reports_to</code> .
Severity	S1 Critical / S2 High / S3 Medium / S4 Low. Used on Incidents, where priority is used on tickets.
SLA	Service Level Agreement — the response and resolution promises, measured in working time.
SLA status	Ongoing, Paused, Fulfilled, Failed, Response Overdue, Resolution Overdue or Not Applicable.
Urgency	How quickly the requester needs it. Combined with impact to derive priority.
Watcher	Someone copied on a ticket without owning it, at a subscription level of their choosing.
Working time	The policy's business hours minus holidays. All SLA arithmetic uses it unless the policy is 24×7.

Turacoz IT Service Desk · version 1.0.1 · module `Turacoz IT Service Desk` in app `turacoz_service_desk`.
Raised and maintained by the Turacoz IT team — it@turacoz.com.